

The University of Faisalabad

Department of Computer Science



The
University of
Faisalabad

Computer Networks
CS-314

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Lab #1: Introduction to Cisco Packet Tracer

This lab introduces the fundamentals of IP addressing, subnetting, and the Cisco Packet Tracer tool – essential skills for any aspiring network engineer.

Lab Outcomes

By the end of this lab, you will be able to:

IP Addressing

Understand what an IP address is and how it acts as a logical address for devices on a network.

IP Address Classes

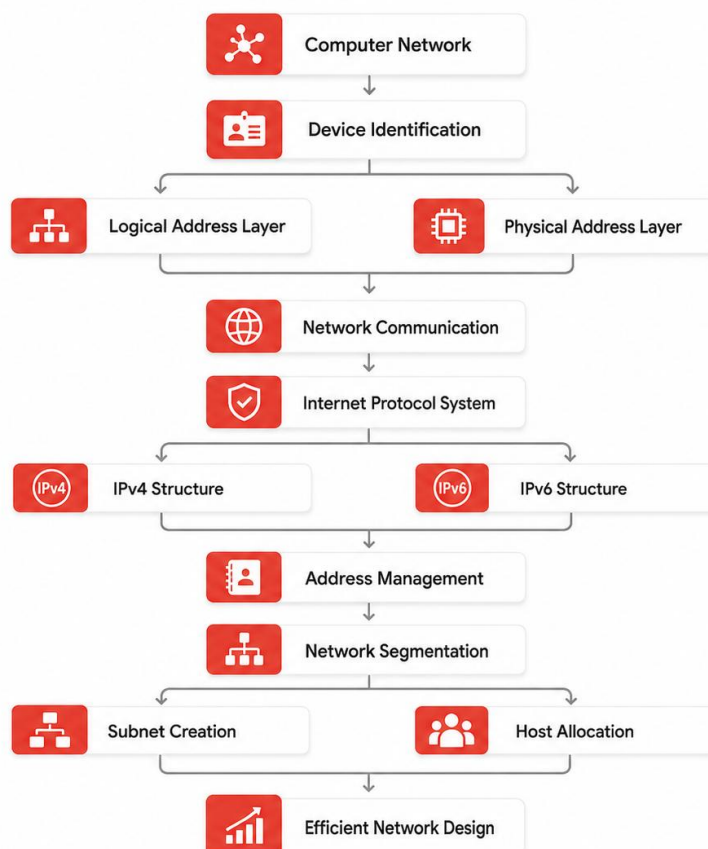
Identify the different classes of IP addresses (Class A, B, C, D, and E) and know which ones are used for different network sizes.

Subnetting

Explain the concept of subnetting and how it divides large networks into smaller, more secure, and efficient pieces.

Cisco Packet Tracer

Successfully download, install, and launch the Cisco Packet Tracer software on your computer.



IP Addresses, Classes & Subnetting

What is an IP Address?

Think of an IP address like a house address, but for your computer or phone. It is a unique number given to any device on a network so they can find and talk to each other.

- **Logical vs. Physical:** An IP address is "logical" (it can change depending on what network you connect to), while a MAC address is "physical" and tied to the device's actual hardware.
- **IPv4:** The standard 32-bit IP address that most devices use today.
- **IPv6:** Because the world is running out of IPv4 addresses, we created IPv6, which is much longer (128-bit) and provides way more addresses.
- **Public IP:** Your global address given by your internet provider, used to talk to the wider internet.
- **Private IP:** A local address used only inside your home or school network, keeping devices hidden from the outside internet.

The Classes of IP Addresses

IP addresses are grouped into "Classes" based on the size of the network. Here is how they are divided:

Class	Range	Capacity
A	1.0.0.0 – 126.255.255.254	Up to 16 million devices
B	128.1.0.1 – 191.255.255.254	Up to 65,000 devices
C	192.0.1.1 – 223.255.254.254	Up to 254 devices
D	224.0.0.0 – 239.255.255.255	Reserved for multicasting
E	240.0.0.0 – 254.255.255.255	Reserved for future / experimental use

Fun Fact: Addresses starting with 127 are reserved for "loopback" – a way for a computer to send a message to itself to test if its own network card is working.

Subnetting & Supernetting

Subnetting is like taking a giant pizza and slicing it into smaller, manageable pieces. In networking, it means dividing a large network into smaller mini-networks to improve speed, security, and efficiency. **Supernetting** is the exact opposite – combining multiple small networks into one large one.

Total Subnets

Calculate this using 2^n , where n is the number of bits borrowed for the network.

Usable Devices (Hosts)

Calculate this using $(2^h) - 2$, where h is the bits left for devices. We subtract 2 because the very first and very last address of a network are always reserved.

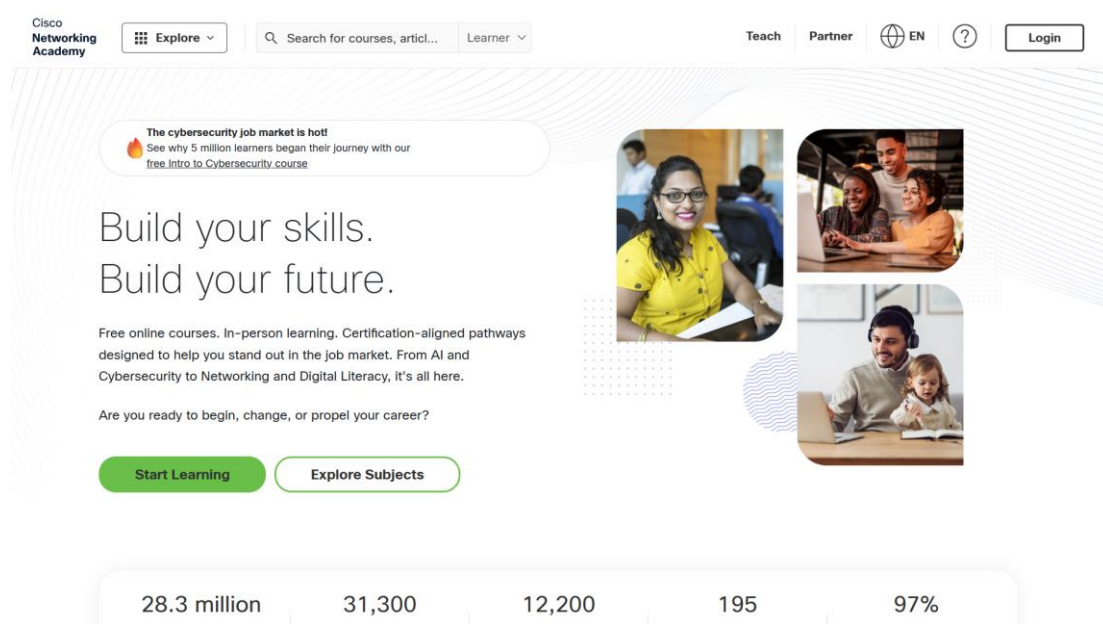
Installing Cisco Packet Tracer

Cisco Packet Tracer is a tool that lets you practice building networks safely on your computer. Follow these steps to get it running:

01

Open your web browser

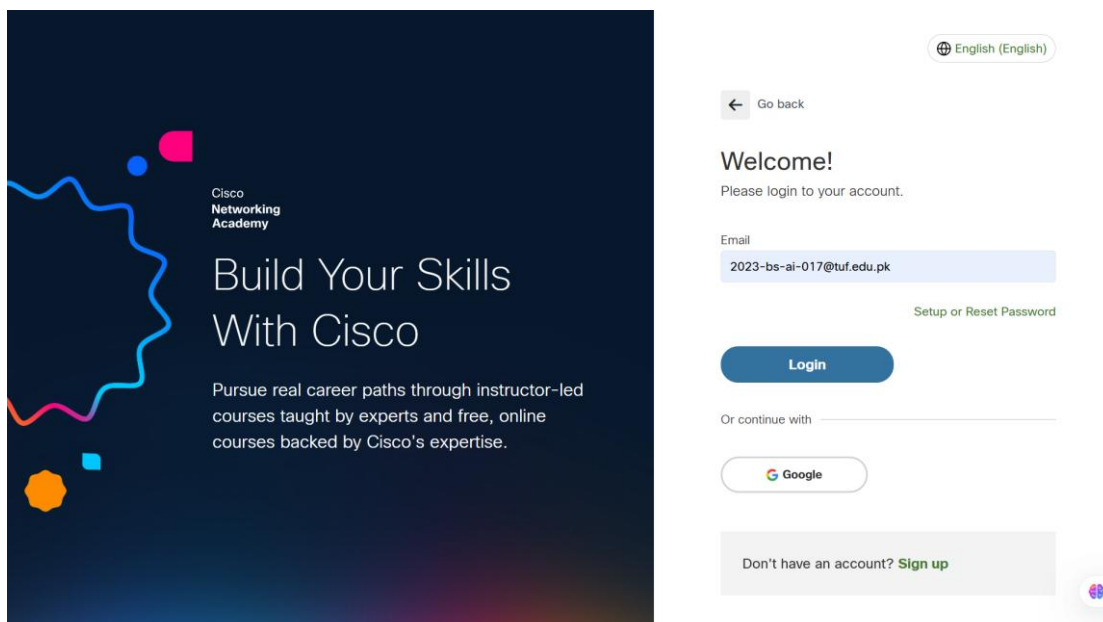
Go to the official Cisco Network Academy download page.



02

Log in

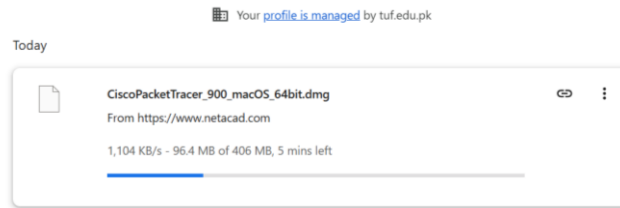
Use your university or school email address.



03

Download the setup file

Download the setup file for your computer.



04

Run the installer

Open the downloaded .exe file and let the installation wizard guide you.



CiscoPacketTracer_900_win_64bit

Type: Application

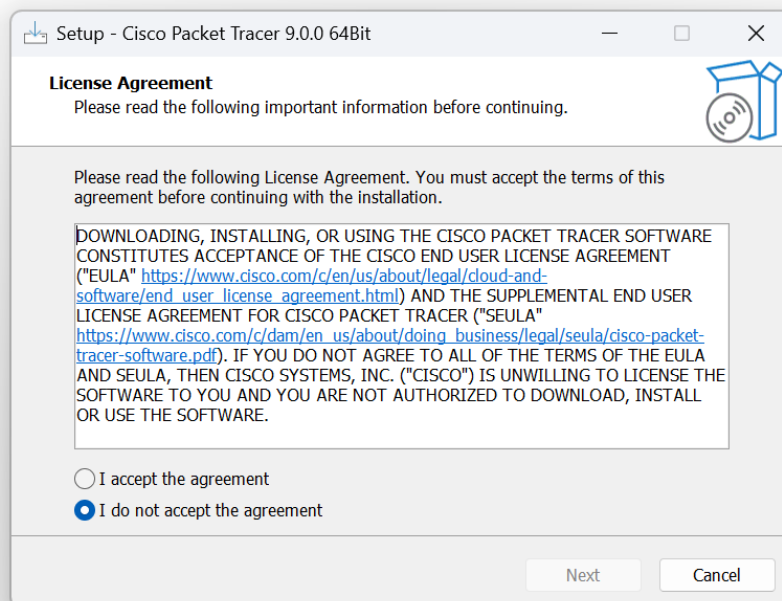
Date modified: 16/05/2026 11:14 pm

Size: 287 MB

05

Accept the License Agreement

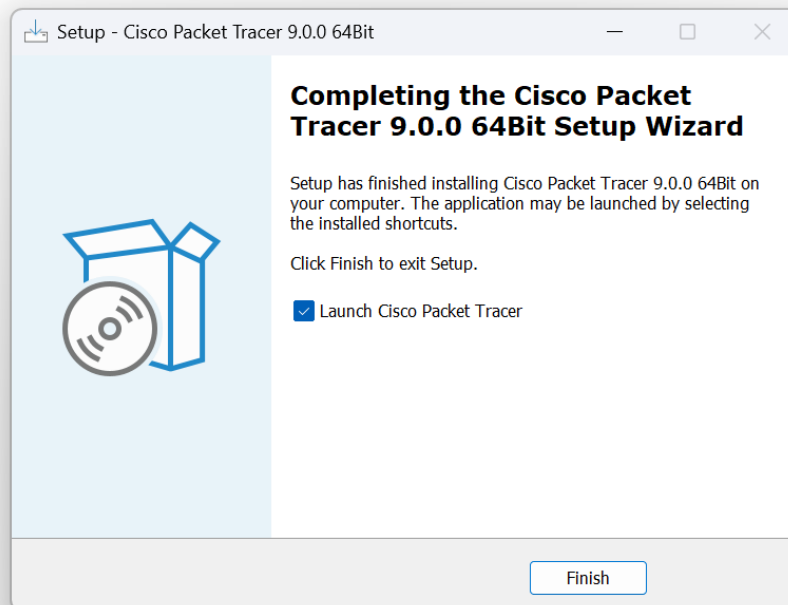
Read and accept the Digital License Agreement when it pops up, then hit install.



06

Finish & Launch

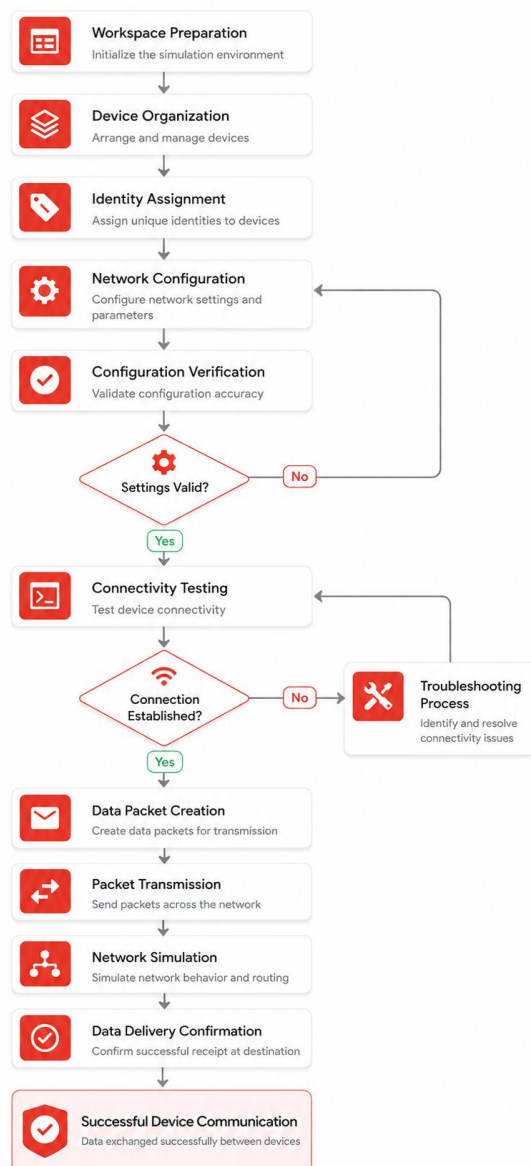
Click "Finish" once it completes. Open the app! You can log in to start your trial. If you want to skip the login screen entirely, just disconnect your computer from the Wi-Fi before opening the app.



 **Lab Work Recap:** This lab covered three parts – (1) Understanding Network Addresses (comparing IP vs MAC addresses, reviewing IPv4 vs IPv6, memorizing IP address classes), (2) Subnetting Calculations (practicing 2^n for subnets and $(2^h)-2$ for usable hosts), and (3) Installing Cisco Packet Tracer (downloading from Cisco Network Academy, running the installer, accepting the license agreement, and launching the app).

Lab # 2: IP Assigning, Commands for CMD, Labeling method, Simple packet Transformation

By the end of this lab, you will be able to use the **Labeling Method** to keep your digital workspace neat and organized, successfully **Assign an IP Address** to a computer using the visual interface, use basic **CMD (Command Prompt) Commands** to check your settings and test connections, and perform a **Simple Packet Transmission** to visually watch data travel from one computer to another.



Lab Work: Step-by-Step

Part 1: The Labeling Method

Before building a big network, it is important to stay organized!

- **Step 1:** Open Cisco Packet Tracer and drag two PCs onto your blank white workspace from the bottom-left menu.
- **Step 2:** Select the **Place Note** tool (it looks like a little sticky note or piece of paper at the top).
- **Step 3:** Click next to your first PC and type a label for it, like "PC 1 - 192.168.1.10". Do the same for the second PC ("PC 2 - 192.168.1.11"). This helps you remember which address belongs to which computer!

Part 2: IP Assigning

Now we have to actually program those addresses into the computers so they can communicate.

- **Step 4:** Click directly on **PC 1**. A window will pop up.
- **Step 5:** Go to the **Desktop** tab at the top of that window, then click on **IP Configuration**.
- **Step 6:** Type 192.168.1.10 into the IPv4 Address box. When you click the "Subnet Mask" box below it, it will automatically fill in 255.255.255.0. Close the window.
- **Step 7:** Repeat this exact process for **PC 2**, but use the address 192.168.1.11.

Part 3: Commands for CMD

Let's make sure the computer knows its own address!

- **Step 8:** Click on **PC 1** again and stay on the **Desktop** tab. This time, click on **Command Prompt**.
- **Step 9:** Type `ipconfig` and hit Enter. This command tells the computer to show you its network settings. You should see the IP address you just typed in!
- **Step 10:** Now, let's test if PC 1 can "hear" PC 2. Type `ping 192.168.1.11` and hit Enter. If you see replies coming back, your computers are ready to talk!

Part 4: Simple Packet Transformation

Now for the best part—watching a message actually travel across the screen!

- **Step 11:** In the bottom right corner of Packet Tracer, click on **Simulation** (it switches you out of "Realtime" mode).
- **Step 12:** At the top of the screen, click the **Add Simple PDU** button (it looks like a closed envelope).
- **Step 13:** Click on **PC 1** (the sender), and then click on **PC 2** (the receiver). You will see a tiny envelope appear on PC 1.
- **Step 14:** Click the **Play** button in the Simulation Panel. Watch the envelope travel from one computer to the other and back! This is exactly how data moves across the internet.

Reference Data

1. The Equipment Data

Before you enter any numbers, you need to drag these items onto your blank workspace.



1x Switch

Go to **Network Devices** > **Switches** and select the **2960** switch. Drag it to the middle of the workspace.

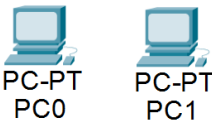


2960-24TT
Switch0



2x PCs

Go to **End Devices**, select **PC**, and drag two of them onto the screen.

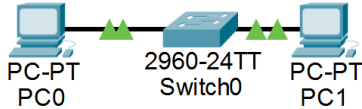


PC-PT
PC0 PC-PT
PC1



Connections

Go to **Connections** (the lightning bolt icon), select the solid black line (**Copper Straight-Through**), and connect PC 1 to the Switch, and PC 2 to the Switch.



PC-PT
PC0 2960-24TT
Switch0 PC-PT
PC1

2. The Labeling Data

Use the **Place Note** tool to add these exact text labels next to your computers so you do not forget their addresses.

Device	Text to Type on the Label
First Computer	PC 1 - 192.168.1.10
Second Computer	PC 2 - 192.168.1.11



3. The IP Configuration Data

Click on each PC, go to the **Desktop** tab, open **IP Configuration**, and type in these exact numbers.

For PC 1

- **IPv4 Address:** 192.168.1.10
- **Subnet Mask:** 255.255.255.0 (This usually fills in automatically when you click the box)

For PC 2

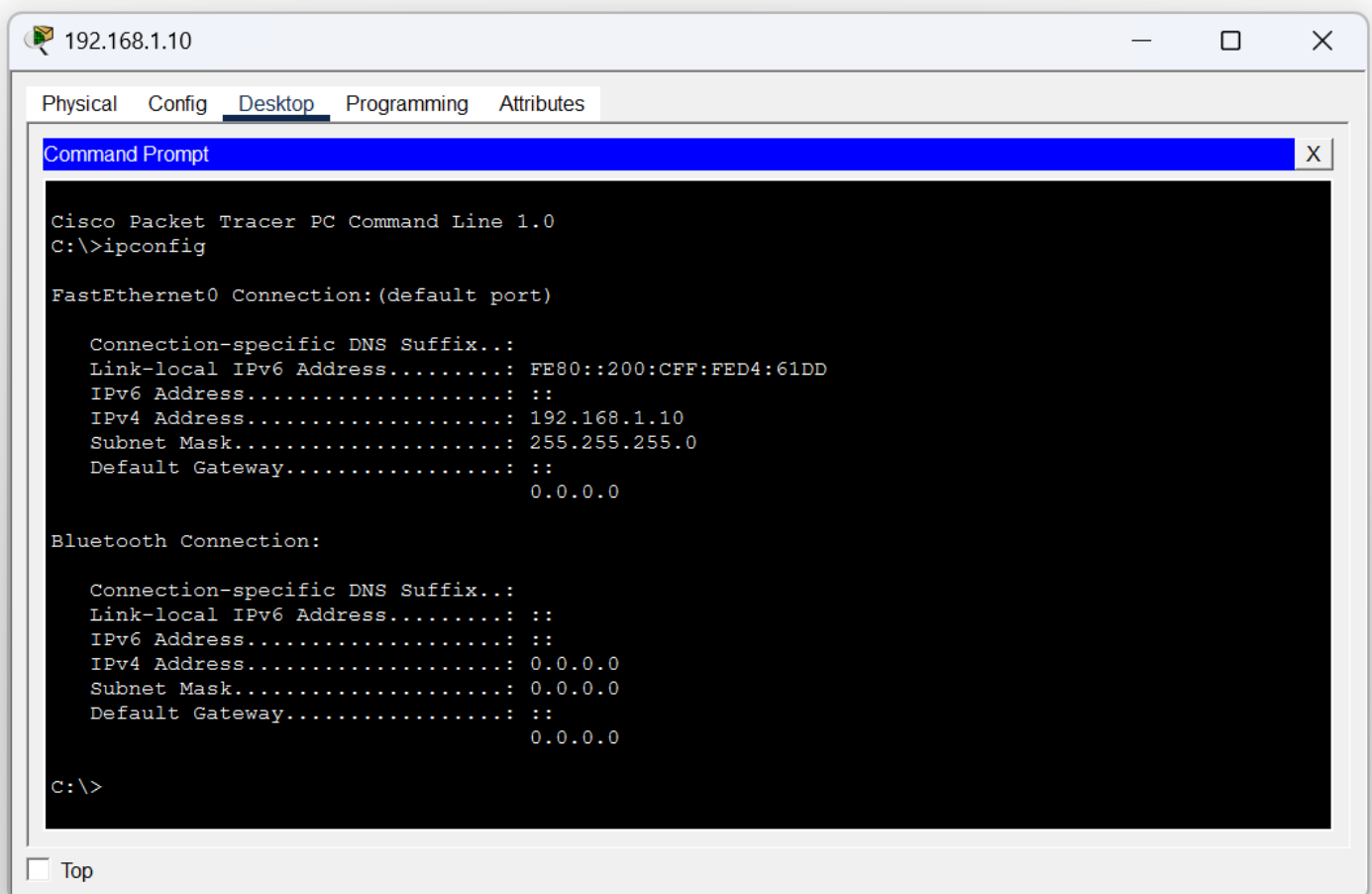
- **IPv4 Address:** 192.168.1.11
- **Subnet Mask:** 255.255.255.0

4. The Command Prompt (CMD) Data

To test if everything is working, click on **PC 1**, go to the **Desktop** tab, and open **Command Prompt**. Type these exact commands and hit Enter after each one.

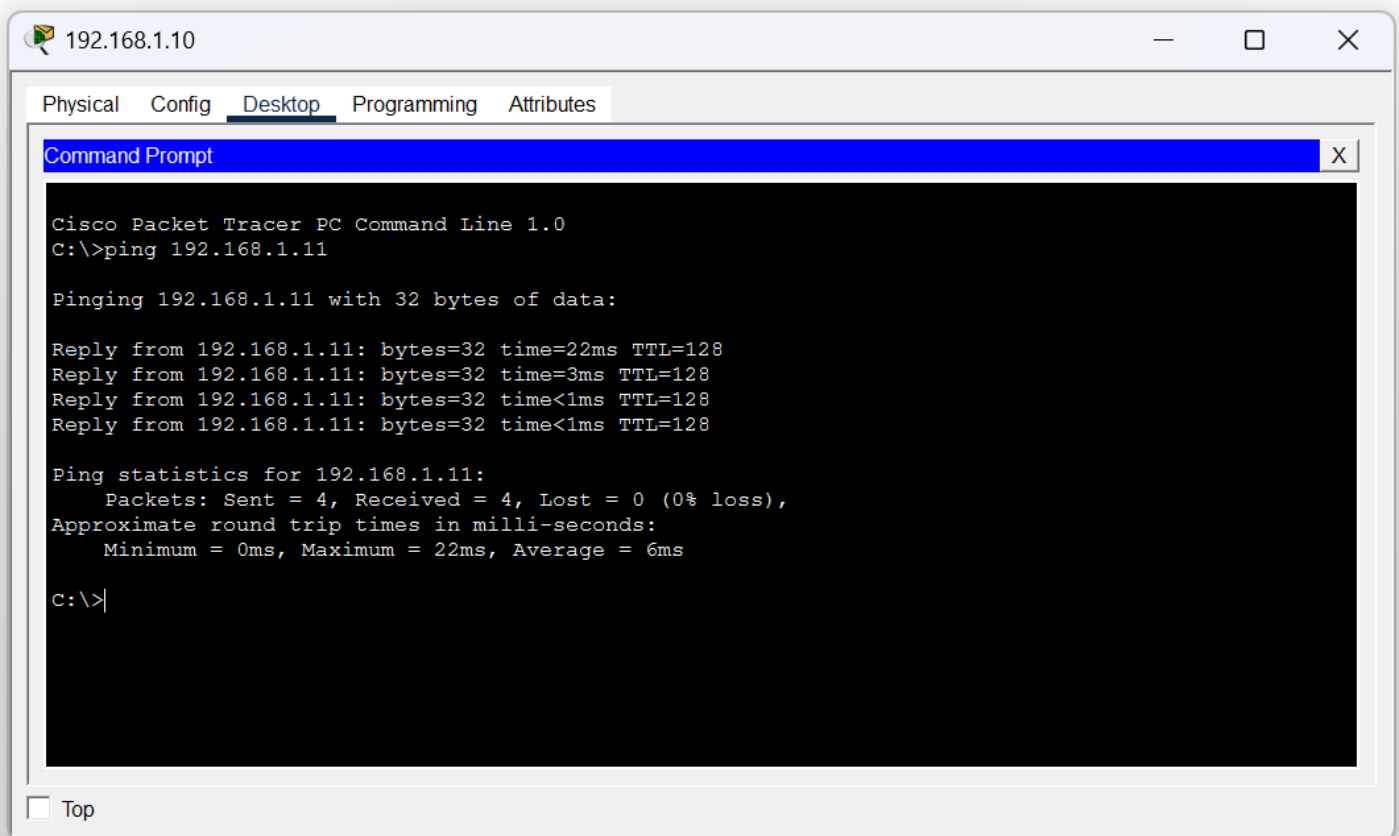
```
ipconfig
```

This will show you the address you just typed in to confirm it saved.



ping 192.168.1.11

This tells PC 1 to send four test messages to PC 2. If it works, you will see a message saying "Reply from 192.168.1.11".



Lab # 3: Constructing a Basic Network Using Two Personal Computers

This hands-on lab walks you through building your very first computer network using Cisco Packet Tracer – no command-line expertise required. Follow each step carefully and you'll have two PCs communicating by the end of the session.

Lab Outcomes

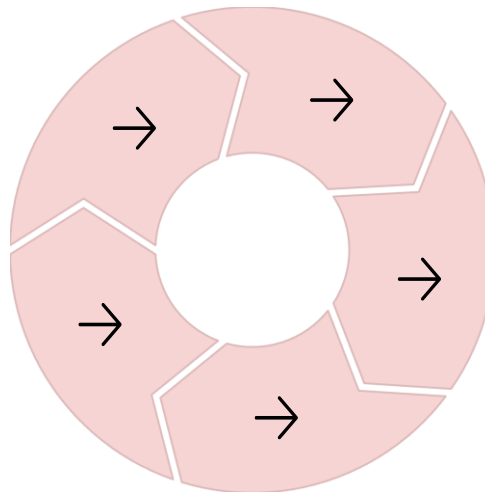
By the end of this lab, you will be able to:

Navigate the Interface

Move confidently around the Cisco Packet Tracer workspace.

Test the Connection

Verify visually that the two computers can communicate with each other.



Place Devices

Add computer devices onto your workspace from the device panel.

Choose the Correct Cable

Select the right physical cable to connect two identical devices.

Assign an IP Address

Give each computer a digital "home address" using visual menus.

Step-by-Step Guide

1

Bring the Computers onto the Screen

Think of the big white screen as your empty desk – let's place two PCs on it.

1. Look at the bottom-left corner. Click the **End Devices** icon (small computer and phone).
2. In the menu next to it, click the first icon labeled **PC**.
3. Click once on the white workspace – **PC0** appears.
4. Repeat to place a second computer – it will be named **PC1**.



2

Connect the Computers with a Cable

Because we are connecting two of the *same* type of device (PC to PC), we must use a **Copper Cross-Over** cable.

1. Click the **Connections** icon (orange lightning bolt) at the bottom-left.
2. Select the black dashed line – hover to confirm it says **Copper Cross-Over**.
3. Click **PC0**, then select **FastEthernet0** from the pop-up menu.
4. Drag to **PC1**, click it, and choose **FastEthernet0** again.
5. Look for a dashed line with **green triangles** at both ends – green means the cable is connected correctly!



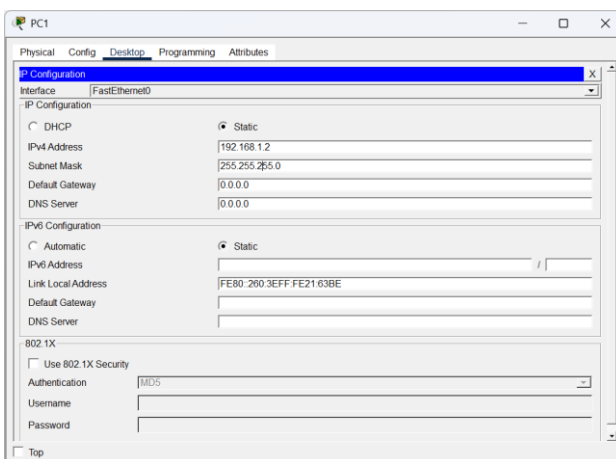
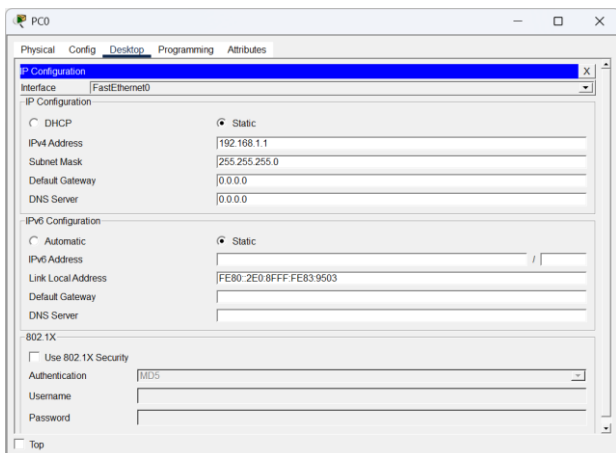
Step-by-Step Guide

3

Give the Computers an IP Address

Computers need an IP address to know who they're talking to, just like houses need street numbers for the mail carrier.

1. Click **PC0** → **Desktop** tab → **IP Configuration**.
2. In **IPv4 Address**, type: **192.168.1.1**
3. Click the **Subnet Mask** box – it auto-fills with **255.255.255.0**. Leave it as is.
4. Close the PC0 window.
5. Repeat for **PC1**: Desktop → IP Configuration → IPv4 Address: **192.168.1.2** (ends in '2' – every computer needs a unique address).
6. Auto-fill Subnet Mask and close the window.



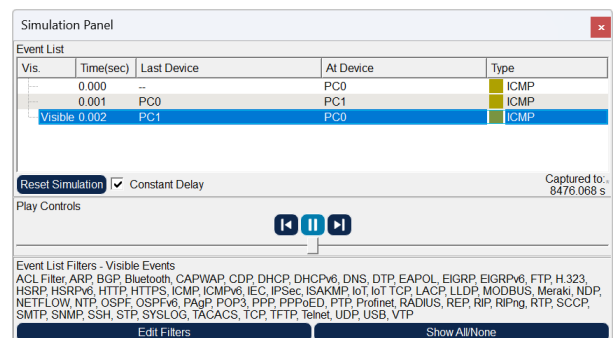
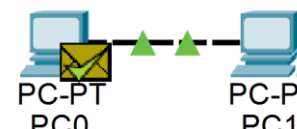
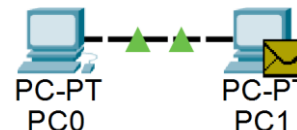
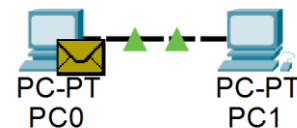
4

Test the Network — Send a Digital Letter

Let's confirm PC0 can successfully send a message to PC1 using the visual envelope tool.

1. Find the **yellow envelope icon** near the top-right toolbar (**Add Simple PDU**). Click it.
2. Your cursor becomes an envelope. Click **PC0** (sender), then **PC1** (receiver).
3. Check the bottom-right corner – a new line should appear with **Successful** under the "Status" column.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
●	Successful	PC0	PC1	ICMP	■	0.000	N	0	(edit)	(delete)



Congratulations! You have successfully built, configured, and tested your very first computer network.

Lab #4: Ethernet Cables, Categories, RJ45 Connectors & Wiring Standards

By the end of this lab, you will understand Ethernet cable structure, cable categories, RJ45 connectors, T568A/B wiring standards, and how to choose the correct cable in Cisco Packet Tracer.

Lab Outcomes

→ Ethernet Cable Structure

Understand what an Ethernet cable is and why the wires inside are twisted.

→ Cable Categories

Explain the difference between cable categories (like Cat5 and Cat6).

→ RJ45 Connector

Identify an RJ45 connector and its purpose.

→ Wiring Standards

Understand the two main color-coding standards (T568A and T568B).

→ Cisco Packet Tracer

Know exactly when to choose a "Straight-Through" or "Cross-Over" cable in Cisco Packet Tracer.



Step 1: Understanding the Ethernet Cable

Think of an Ethernet cable as a **physical highway for digital data**. It connects computers, routers, and switches together.

8 Tiny Wires

If you were to cut open an Ethernet cable, you would not see one big wire. Instead, you would see **8 tiny, color-coded wires**.

4 Twisted Pairs

These 8 wires are twisted together in pairs (4 pairs total).

Why Twist Them?

Twisting the wires acts like an invisible shield. It stops outside electrical noise (like from lights or microwaves) from messing up the data traveling inside. This is called a **"Twisted Pair"** cable.

Step 2: Cable Categories (Cat)

When you buy these cables, they have names like "Cat5e" or "Cat6." "Cat" is just short for Category. Think of these categories like **speed limits on a highway**:

Cat5

An older, slower highway. Rarely used anymore.

Cat5e

The "e" stands for enhanced. It is a standard highway, good for normal internet speeds (up to 1 Gigabit per second).

Cat6

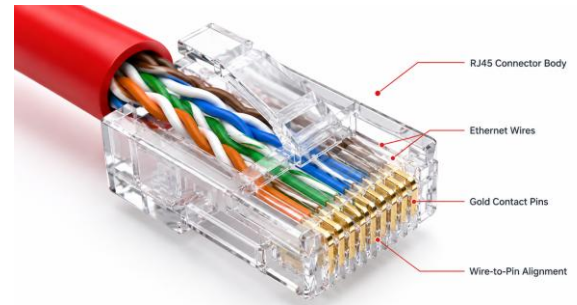
A superhighway. The wires inside are twisted tighter and have better shielding, allowing for much faster data speeds.

RJ45 Connectors & Wiring Standards

Step 3: The RJ45 Connector

To plug this cable into a computer, the 8 tiny wires need a plug. This plastic plug is called an **RJ45 connector**.

- It looks exactly like a clear plastic landline phone plug, but slightly wider.
- If you look closely at an RJ45 connector, it has 8 tiny gold pins. Each of the 8 colored wires inside the cable matches up perfectly with one of these gold pins.



Step 4: Wiring Standards (The Color Codes)

Because there are 8 colored wires (Orange, Green, Blue, Brown, and striped versions of each), we need a universal rule for what order to put them in before we crimp the plastic RJ45 plug onto them. Engineers created two standard color patterns:

1

T568A Pattern

Starts with the Green-Striped wire.

2

T568B Pattern

Starts with the Orange-Striped wire. (This is the most common one used today).

Note for the student: You don't have to memorize the exact color order right now, just know that A and B are the two accepted "recipes" for arranging the wires.

Step 5: Applying This in Cisco Packet Tracer

In Packet Tracer, you don't have to strip wires or use plastic plugs! You just have to click the right cable. Here is how these physical rules apply to your software GUI:

The Straight-Through Cable (**Solid Black Line** in PT)

- **Physical Rule:** Both ends of the cable use the exact same color recipe (T568A on one end, and T568A on the other end).
- **Packet Tracer Rule:** Use this solid black line to connect **DIFFERENT** types of devices. (Example: A PC to a Switch, or a Switch to a Router).

The Cross-Over Cable (**Dashed Black Line** in PT)

- **Physical Rule:** The cable uses different recipes on each end (T568A on one end, and T568B on the other end). This "crosses" the sending and receiving wires so the devices don't talk over each other.
- **Packet Tracer Rule:** Use this dashed black line to connect the **SAME** types of devices. (Example: A PC to a PC, or a Switch to a Switch). *You actually used this cable in Lab 3!*

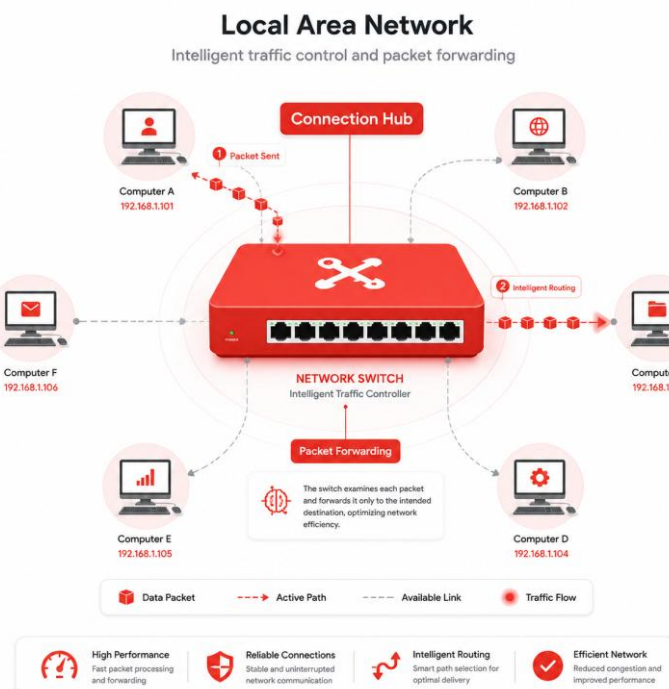
CISCO PACKET TRACER

Lab #5: Making a Local Area Network (LAN) Using a Switch

In this lab, you will build a fully functioning Local Area Network by connecting three computers to a network switch using straight-through cables, assigning IP addresses, and testing communication between all devices.

By the end of this lab, you will be able to:

- Understand the purpose of a Network Switch
- Place a Switch onto the Cisco Packet Tracer workspace
- Apply the rule from Lab 4 to choose the correct cable (Straight-Through)
- Connect multiple computers to form a Local Area Network (LAN)
- Test the network to ensure any computer can talk to any other computer



Lab Outcomes & Step-by-Step Guide



Step 1: Bring the Devices onto the Screen

Let's build a network with three computers and one switch.

1. Look at the bottom-left corner. Click on **End Devices** (the computer/phone icon).
2. Click on the **PC** icon and place **three computers** onto your white workspace. (They will be named PC0, PC1, and PC2).
3. Now, let's get the Switch. Go back to the bottom-left corner and click on **Network Devices** (it looks like a router/circle with arrows).
4. In the menu right next to it, click on **Switches** (it looks like a little rectangular box).
5. Choose the very first switch in the list (it is named **2960**). Click it, and then click in the middle of your workspace to place it.

Step 2: Wire the Network Together

Since a Computer and a Switch are *different* types of devices, remember your rule from Lab 4: we must use a **Straight-Through** cable!

1. Click on the **Connections** icon (the orange lightning bolt).
2. Look for the **solid black line** (Copper Straight-Through) and click it.
3. Click on **PC0** and select **FastEthernet0**.
4. Drag your mouse to the **Switch** and click on it. A long list will pop up! This is because a switch has lots of ports. Click on the very first one: **FastEthernet0/1**.
5. Repeat this for the other computers:
 - Grab another solid black cable. Click **PC1** (FastEthernet0) and connect it to the Switch (**FastEthernet0/2**).
 - Grab a third solid black cable. Click **PC2** (FastEthernet0) and connect it to the Switch (**FastEthernet0/3**).
6. **Important Wait Time:** You will notice the little triangles on the switch side of the cables are **orange**. A real switch takes about 30 seconds to turn on and check the cables. Wait until those orange dots turn **green**.

IP Configuration & Testing Your LAN

1

Step 3: Give the Computers their IP Addresses

They are physically plugged in, but they still need their digital street addresses to talk to each other.

1. Click on **PC0**. Go to the **Desktop** tab -> **IP Configuration**.
2. Type in the IPv4 Address: **192.168.1.1** and click the Subnet Mask box so it auto-fills (**255.255.255.0**). Close the window.
3. Click on **PC1**. Go to **Desktop** -> **IP Configuration**.
4. Type in the IPv4 Address: **192.168.1.2** and click the Subnet Mask box. Close the window.
5. Click on **PC2**. Go to **Desktop** -> **IP Configuration**.
6. Type in the IPv4 Address: **192.168.1.3** and click the Subnet Mask box. Close the window.

2

Step 4: Test Your New LAN!

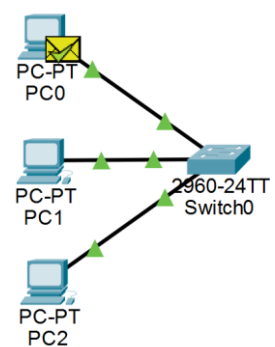
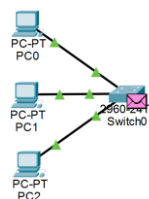
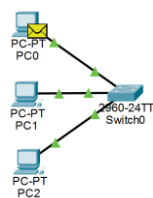
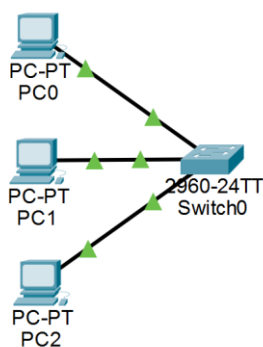
Let's make sure the switch is doing its job as the middleman.

1. Click the **closed yellow envelope** icon (Add Simple PDU) near the top-right toolbar.
2. Click on **PC0** (the sender).
3. Click on **PC2** (the receiver).
4. Look at the bottom-right corner in the status window. It should say **Successful**!
5. Try it again! Grab a new envelope, click on **PC2**, and send it to **PC1**. It should also be Successful.



Congratulations! You have just built a fully functioning Local Area Network (LAN). All the computers can now safely share data with each other through the Switch.

Visualization:



Lab # 6: Repeaters & Routers with Switches & Hubs

As networks get bigger, we need different tools to handle the traffic. In this lab, we are going to build **two separate towns** (networks) and connect them.

First, here is a quick cheat sheet on the devices we are using so a 10th-grade student can easily understand them:



Hub — The Megaphone

An older device. If one computer sends a message, the Hub shouts it to *every* computer plugged into it. It is noisy and not very smart.



Switch — The Smart Mailman

You used this in Lab 5! It looks at the address on the message and delivers it *only* to the correct computer.



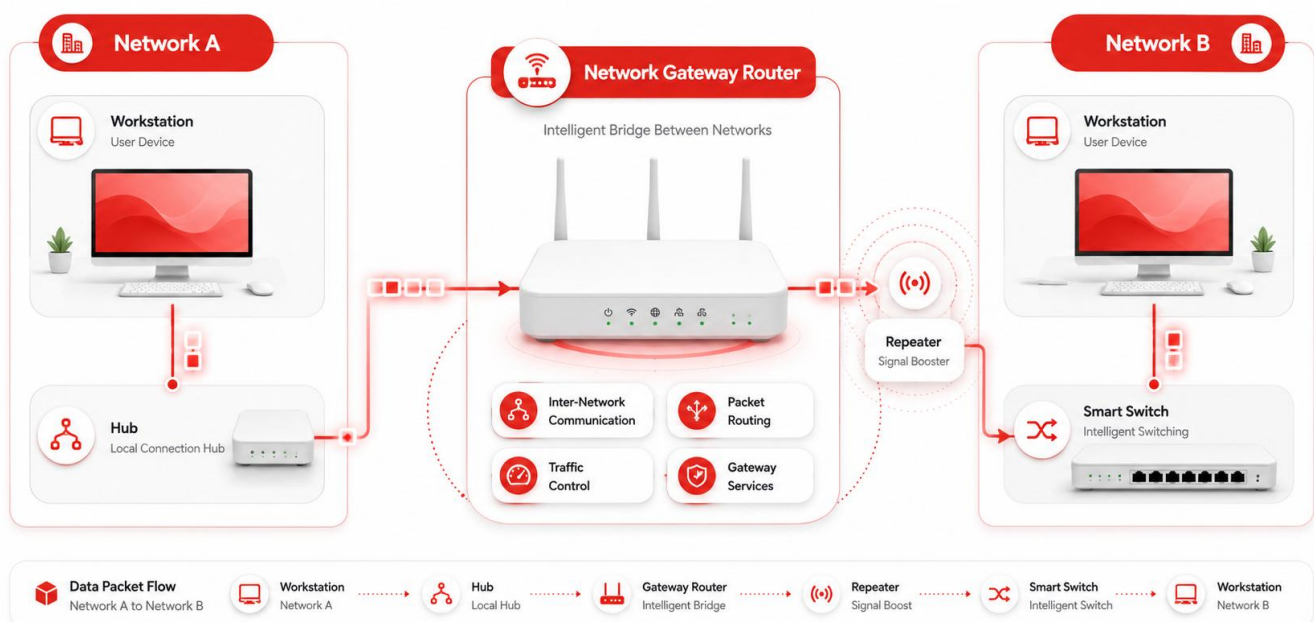
Repeater — The Energy Drink

If a cable is too long, the digital signal gets tired and weak. A repeater takes the weak signal, boosts it back to 100%, and sends it on its way.



Router — The Border Gate

Switches and Hubs can only connect computers in the *same* town. A Router connects *different* towns together so they can share data.



Lab Outcomes & Step 1: Place the Devices

By the end of this lab, you will be able to:

- Understand the difference between a Hub, Switch, Repeater, and Router
- Use the "Auto-Connect" magic cable tool in Packet Tracer
- Configure a Router using the visual GUI (without typing complex code)
- Set up a "Default Gateway" on a computer so it knows how to send messages to a different network

Step 1: Place the Devices on Your Workspace

We are going to build "**Town A**" on the left, "**Town B**" on the right, and put a Router in the middle to connect them.

1. **The Computers:** Go to **End Devices** and place one **PC** on the far left (PC0) and one **PC** on the far right (PC1).
2. **The Hub (Town A):** Go to **Network Devices** → **Hubs** (next to the switches) → click **PT-Hub** and place it next to PC0.
3. **The Switch (Town B):** Go to **Network Devices** → **Switches** → click the **2960** switch and place it next to PC1.
4. **The Repeater (Town B):** Go to **Network Devices** → **Wireless Devices** (or sometimes found under Hubs depending on the version). Look for **PT-Repeater** and place it between your Switch and the middle of the screen.
5. **The Router (The Middle):** Go to **Network Devices** → **Routers** → click the first one named **PT-Router** (this is a generic, beginner-friendly router). Place it dead in the center of your screen.



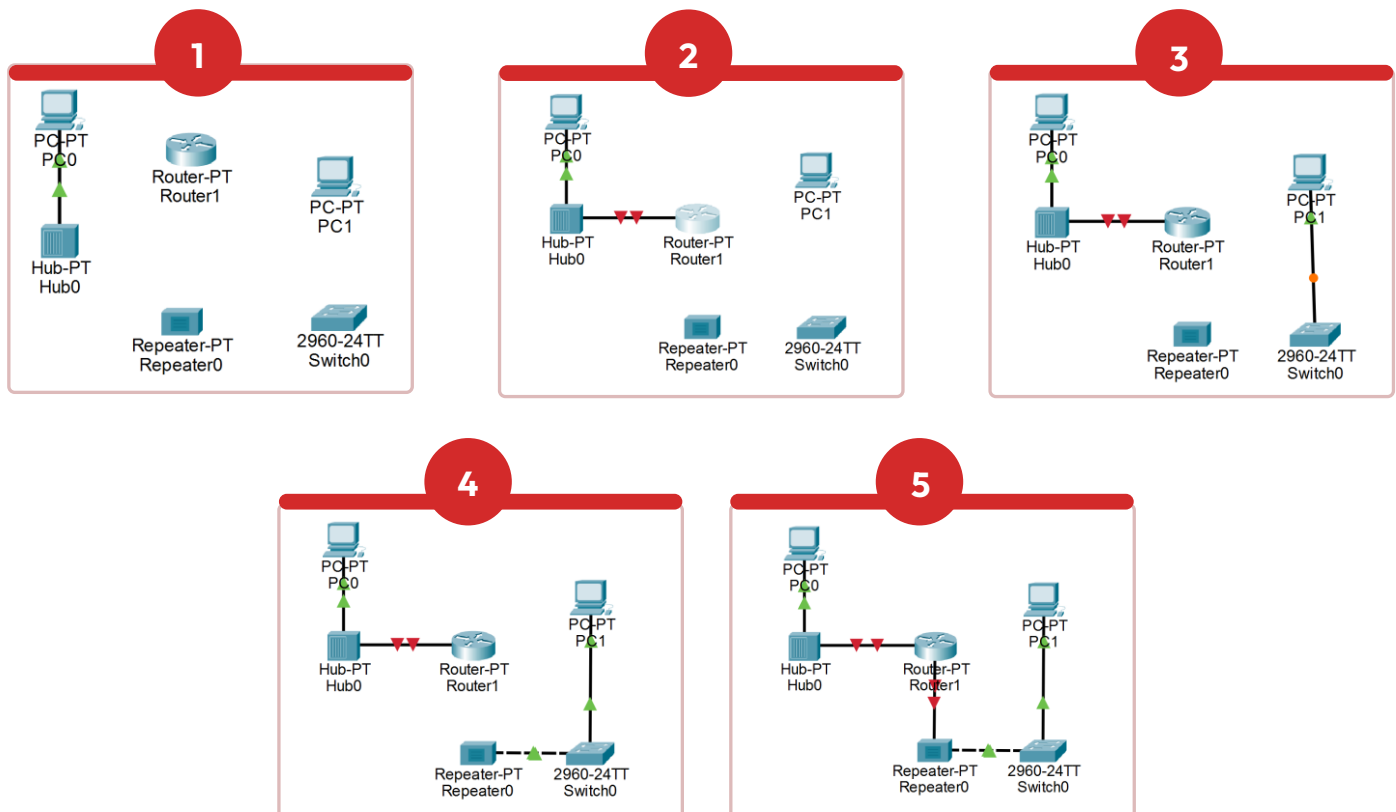
Step 2: Wire it All Together (The Magic Cable)

Because we have so many different devices, remembering which cable to use (Straight-Through vs. Cross-Over) can be tricky. Let's learn a new trick!

1. Click the **Connections** icon (the orange lightning bolt).
2. In the cable menu, click the **first icon** (it looks like a yellow lightning bolt with sparkles). This is the "Auto Choose" tool. It will pick the perfect cable for you automatically!
3. Click **PC0** and then click the **Hub**. (Town A is wired).
4. Click the **Hub** and then click the **Router**. (Town A is connected to the gate).
5. Click **PC1** and then click the **Switch**.
6. Click the **Switch** and then click the **Repeater**.
7. Click the **Repeater** and then click the **Router**. (Town B is wired and connected to the gate).



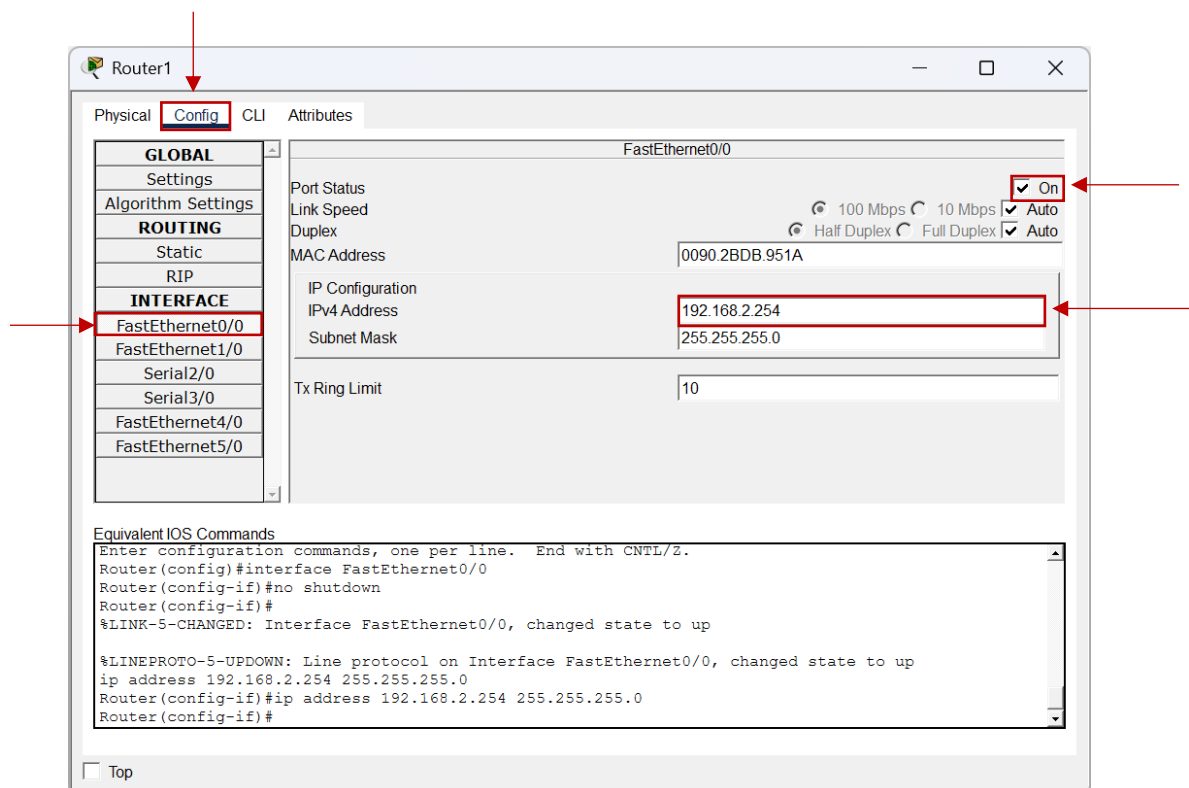
Note: Wait for the little orange circles on the switch to turn green before moving on!



Step 3: Turn on the Router (The Border Gate)

Right now, the connections touching the Router have red triangles. That means the gates are closed. Let's open them!

1. Click on the **Router**. A window will pop up.
2. At the top, click on the **Config** tab.
3. Look at the menu on the left side. Under the word "INTERFACE", click on **FastEthernet0/0** (this is the gate pointing to Town A).
4. Look for the box that says **Port Status** and check the box next to **On**. (You will see the red triangle turn green on your workspace!).
5. In the **IPv4 Address** box, type Town A's gate address: **192.168.1.254**
6. Click the **Subnet Mask** box so it auto-fills, then go back to the left menu.
7. Click on **FastEthernet0/1** (this is the gate pointing to Town B).
8. Check the **Port Status: On** box.
9. In the **IPv4 Address** box, type Town B's gate address: **192.168.2.254**
10. Click the Subnet Mask box so it auto-fills. Close the Router window.



Step 4: Configure the Computers & Step 5: Test the Connection

Step 4: Configure the Computers

The computers need their own IP addresses, but they **ALSO** need to know the address of the Router so they can send mail out of their town. This is called the **Default Gateway**.

PC0 (Town A):

1. Click **PC0** (in Town A). Go to **Desktop** → **IP Configuration**.
2. IPv4 Address: **192.168.1.1**
3. Click Subnet Mask so it auto-fills.
4. **Default Gateway**: Type in Town A's gate address: **192.168.1.254**. Close the window.

PC1 (Town B):

1. Click **PC1** (in Town B). Go to **Desktop** → **IP Configuration**.
2. IPv4 Address: **192.168.2.1** (*Notice it is a '2', because it is a different town!*)
3. Click Subnet Mask so it auto-fills.
4. **Default Gateway**: Type in Town B's gate address: **192.168.2.254**. Close the window.

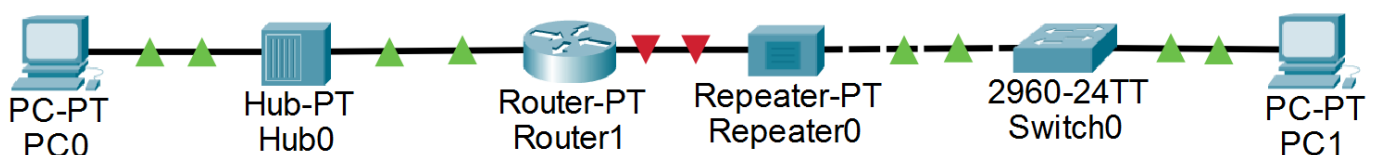
Step 5: Test the Long-Distance Connection!

1. Grab the **closed yellow envelope** (Add Simple PDU) from the top toolbar.
2. Click on **PC0** (Sender in Town A).
3. Click on **PC1** (Receiver in Town B).
4. Look at the bottom-right status window.



Important Tip: The very first time a router tries to find a new town, it drops the first package while it draws a map. **If it says "Failed", do not panic!** Just click the "Fire" button (double-click the red "Failed" text, or just send a brand new envelope from PC0 to PC1).

The second attempt will say **Successful**. Your towns are communicating!



LAB #7

Lab # 7 : Router Configuration

In this lab, you will configure a router to connect two separate networks and verify that it can successfully pass messages between them using Cisco Packet Tracer.

Lab Outcomes

Router Fundamentals

Understand the main job of a Network Router.

Naming & IP Configuration

Change the name of a Router using the graphical interface and configure IP addresses on different Router ports (interfaces).

Default Gateway

Understand why the Router acts as the "Default Gateway" for computers.

Verification

Prove that your configured Router can successfully pass messages between two different networks.

Lab Work: Building the Network

Step 1: Build the Two Networks

Place **two PCs** (PC0 left, PC1 right), a **2960 Switch** next to each, and the **PT-Router** in the middle. Use the **Magic Auto-Connect Cable** (yellow lightning bolt) to wire them up:

- PC0 → Switch0
- Switch0 → Router
- PC1 → Switch1
- Switch1 → Router



Step 2: Give the Router a Name

Click the **Router** → **Config** tab → **Settings** under "GLOBAL". Change both **Display Name** and **Hostname** to **Main_Post_Office**. (Notice the underscore – network names don't like spaces!)

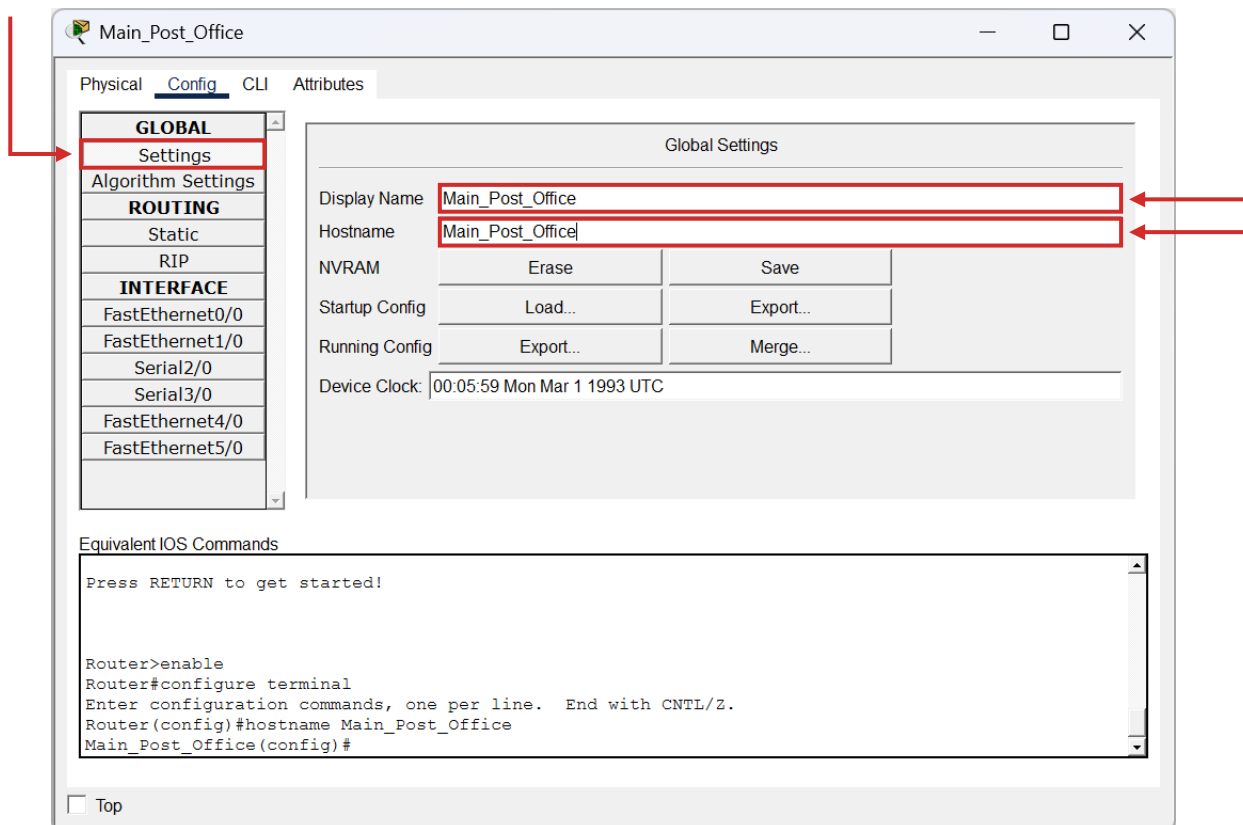
Step 3: Configure the Router's Ports

Stay on the **Config** tab. Under "INTERFACE":

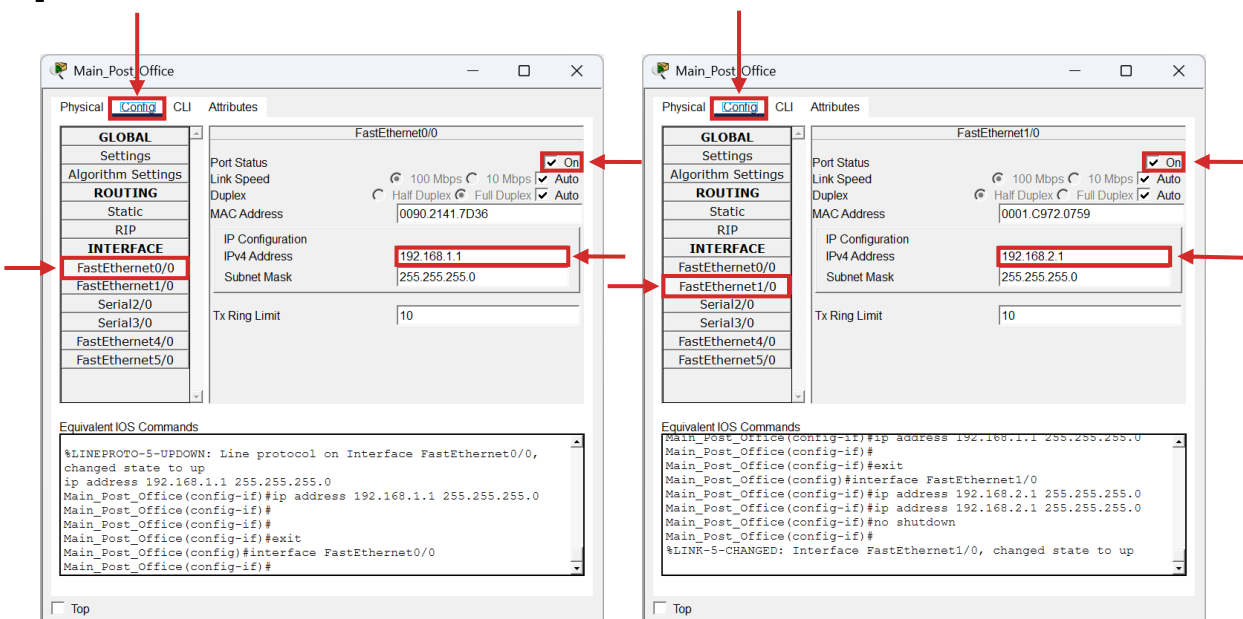
- **FastEthernet0/0** (left network): Port Status **On**, IPv4 **192.168.1.1**
- **FastEthernet0/1** (right network): Port Status **On**, IPv4 **192.168.2.1**

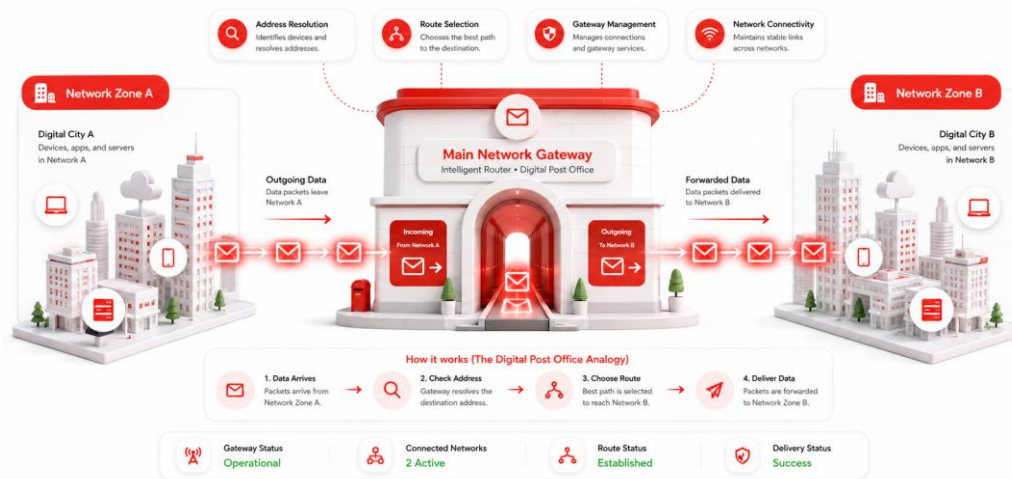
Red triangles will turn **green** once ports are active.

Step 2 Visualization



Step 3 Visualization



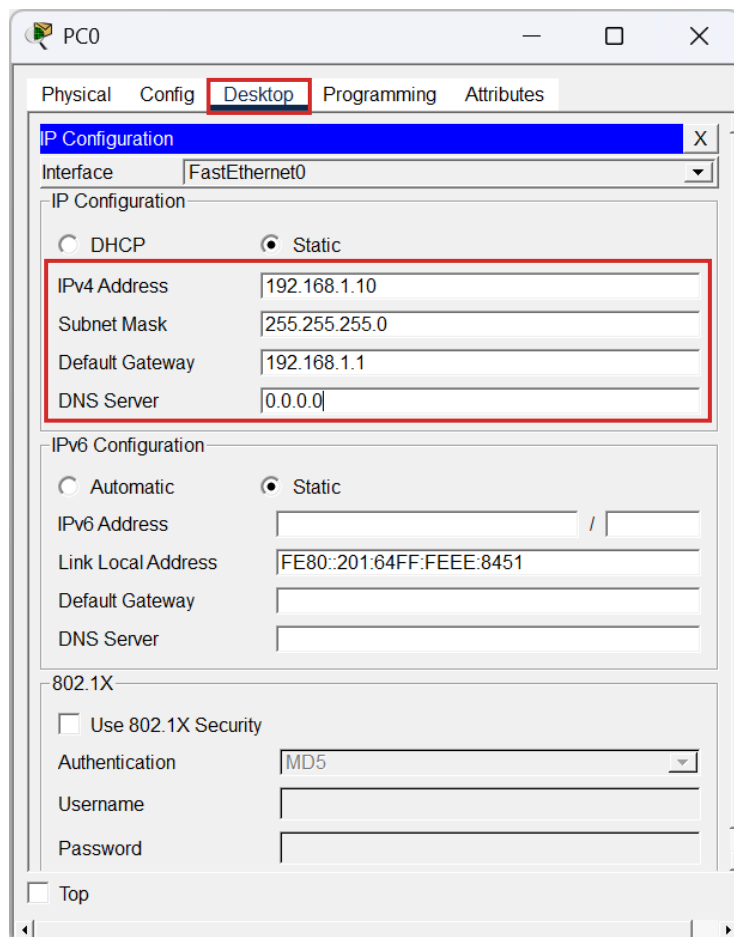


Configuring PCs & Testing the Router

1

Step 4: Set the Default Gateway on PC0

Click PC0 → Desktop → IP Configuration. Set IPv4 Address to 192.168.1.10, click the Subnet Mask box, and set **Default Gateway** to 192.168.1.1 (the router's left-side address).

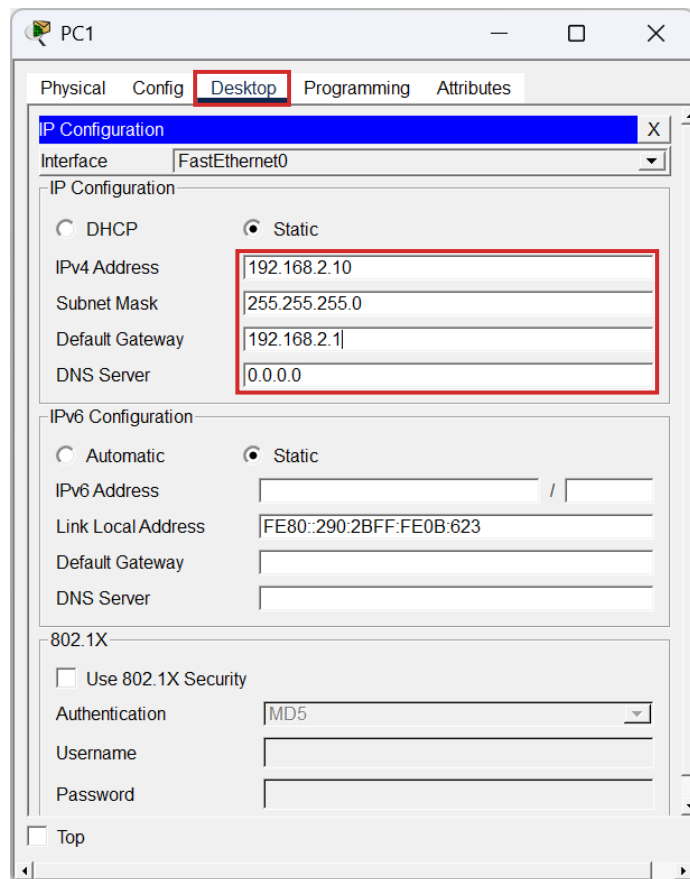


Tip: The router must build its routing table the very first time a message crosses networks. That is why the initial ping or envelope always fails – simply retry and it will succeed.

2

Step 4 (continued): Set the Default Gateway on PC1

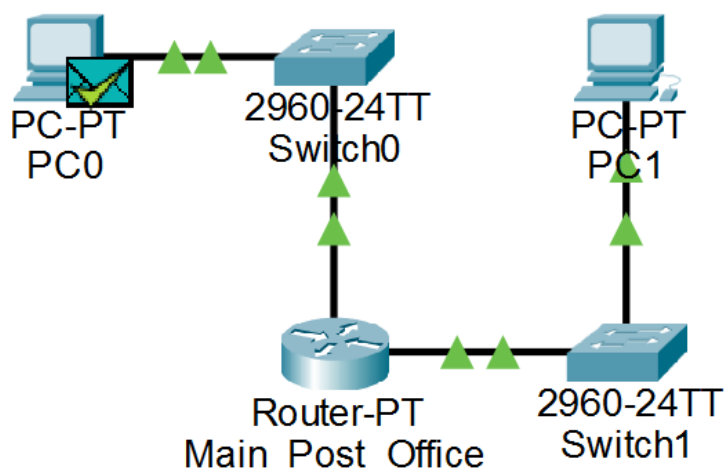
Click PC1 → Desktop → IP Configuration. Set IPv4 Address to 192.168.2.10, click the Subnet Mask box, and set **Default Gateway** to 192.168.2.1 (the router's right-side address).



3

Step 5: Test the Router!

Grab the **closed yellow envelope** from the toolbar. Click PC0 then PC1. The *first* message will say **Failed** – the router is drawing its map. Double-click the red text or send a new envelope. The **second time** it will say **Successful!** 🎉

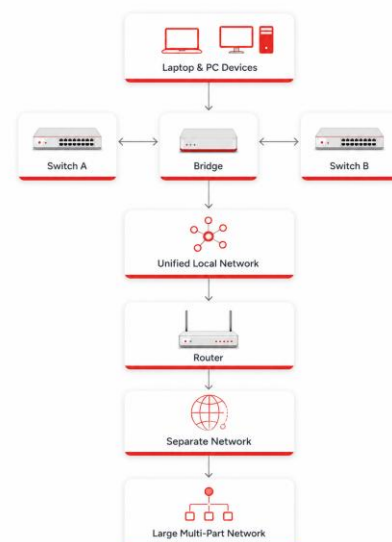


Lab #8: Building an Advanced Network

Routers, Switches, Bridges, PCs & Laptops – using Cisco Packet Tracer

By the end of this lab, you will be able to:

- Add Laptops to your Packet Tracer workspace and configure them.
- Understand the difference between a Bridge (connects the *same* network) and a Router (connects *different* networks).
- Use a Network Bridge to connect two Switches together.
- Build a large, multi-part network using the graphical interface.



Lab Work: Step-by-Step Guide

Step 1: Place the Devices – Building Our Two Towns

We are going to build **Town A** (which is so big it needs two switches and a bridge) and **Town B**. Then, we will put a **Router** in the middle to connect the two towns.

1

The Middle (The Gate)

Go to **Routers** and place a **PT-Router** in the top center of your screen.

2

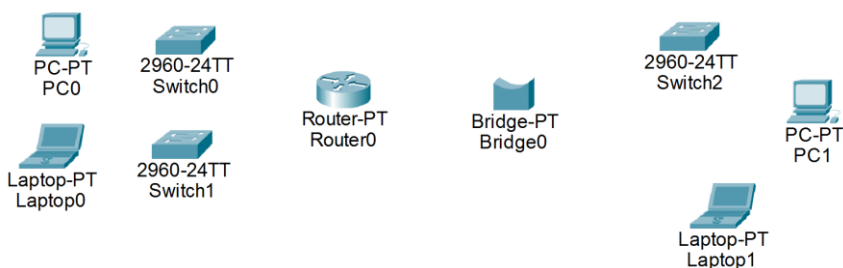
Town A (Left Side)

- Go to **Switches** and place **two 2960 Switches** on the left side of your screen.
- Go to **Network Devices** → **Switches**, look for the **PT-Bridge** (it looks like a small box with two arrows), and place it right between your two switches.
- Go to **End Devices**. Place one **PC** on the first switch, and one **Laptop** on the second switch.

3

Town B (Right Side)

- Place **one 2960 Switch** on the right side of your screen.
- Go to **End Devices** and place one **PC** and one **Laptop** near this switch.



Step 2: Wire It All Together — Using the Magic Cable

Because we have a lot of devices, we will use the "Auto Choose" cable to prevent mistakes.

- Click the **Connections** icon (the orange lightning bolt). Click the **first icon** (the yellow lightning bolt with sparkles).

Wire Town A

- Click the PC and connect it to the first Switch.
- Click the first Switch and connect it to the **Bridge**.
- Click the **Bridge** and connect it to the second Switch.
- Click the Laptop and connect it to the second Switch.

Wire Town B

- Connect the PC to Town B's Switch.
- Connect the Laptop to Town B's Switch.

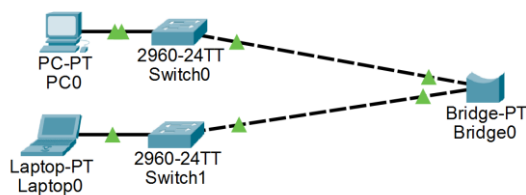


Connect to the Router

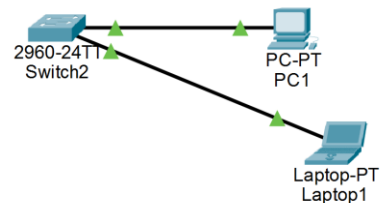
- Connect Town A's *first* Switch to the **Router**.
- Connect Town B's Switch to the **Router**.

Wait for the orange dots on the switches to turn green!

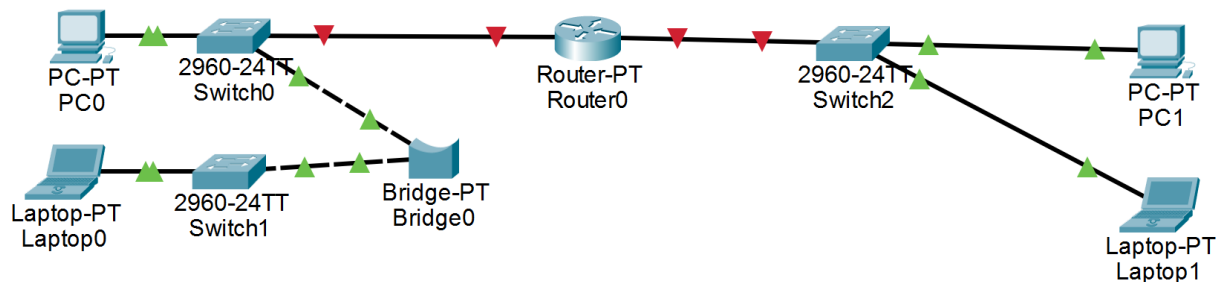
1



2



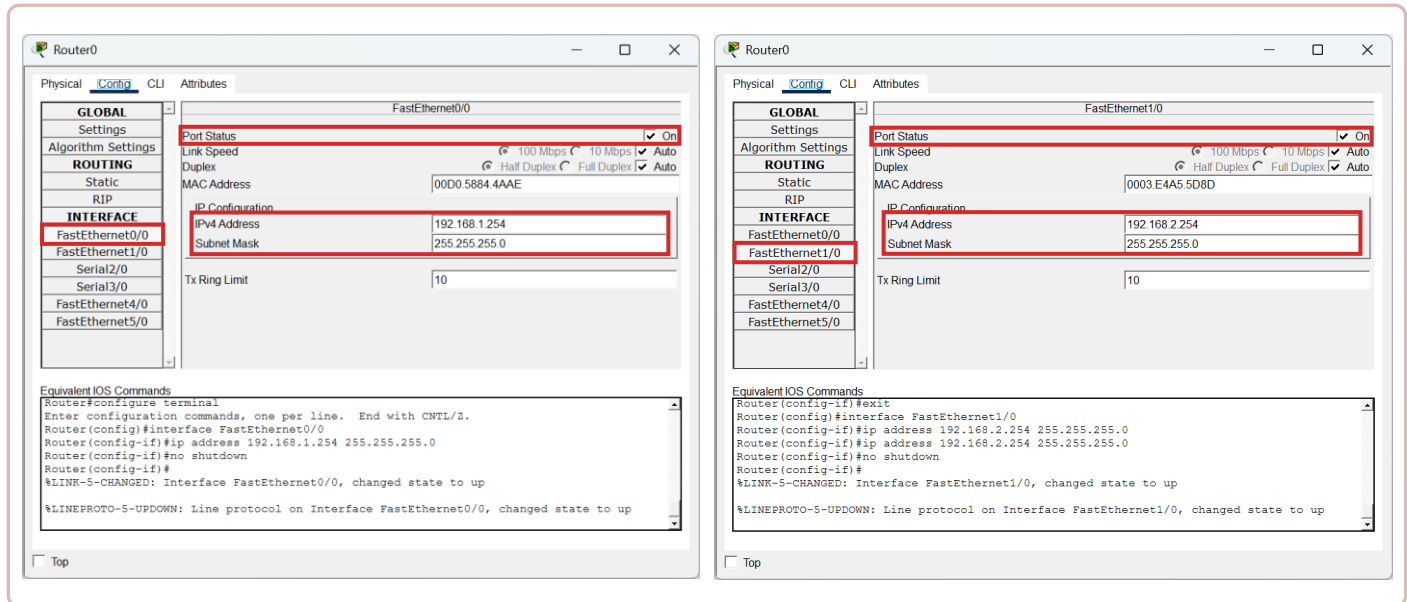
3



Configuration & Testing

Step 3: Turn on the Router (The Main Post Office)

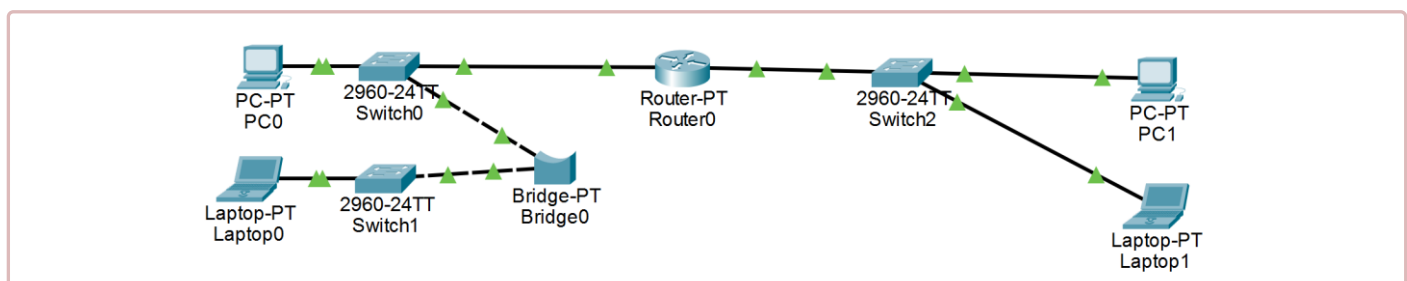
1. Click on the **Router** → go to the **Config** tab.
2. On the left menu, click **FastEthernet0/0** (pointing to Town A).
3. Check **Port Status: On**. Set IPv4 Address to: **192.168.1.254** (Click Subnet Mask to auto-fill).
4. On the left menu, click **FastEthernet0/1** (pointing to Town B).
5. Check **Port Status: On**. Set IPv4 Address to: **192.168.2.254** (Click Subnet Mask to auto-fill). Close the window.



Step 4: Configure the Computers and Laptops

Laptops work exactly like PCs in Packet Tracer! Because Town A has a Bridge, *both* switches in Town A share the exact same IP neighborhood.

Device	IPv4 Address	Subnet	Default Gateway	Network
Town A – PC	192.168.1.1	Auto-fill	192.168.1.254	Town A (Bridge)
Town A – Laptop	192.168.1.2	Auto-fill	192.168.1.254	Town A (Bridge)
Town B – PC	192.168.2.1	Auto-fill	192.168.2.254	Town B
Town B – Laptop	192.168.2.2	Auto-fill	192.168.2.254	Town B



Notice: Town A devices use **.1.x** addresses because the Bridge keeps them in the same town. Town B devices use **.2.x** because they are in a different network!

Step 5: Test the Entire Network!

Let's make sure the Bridge and the Router are doing their jobs.

Test the Bridge

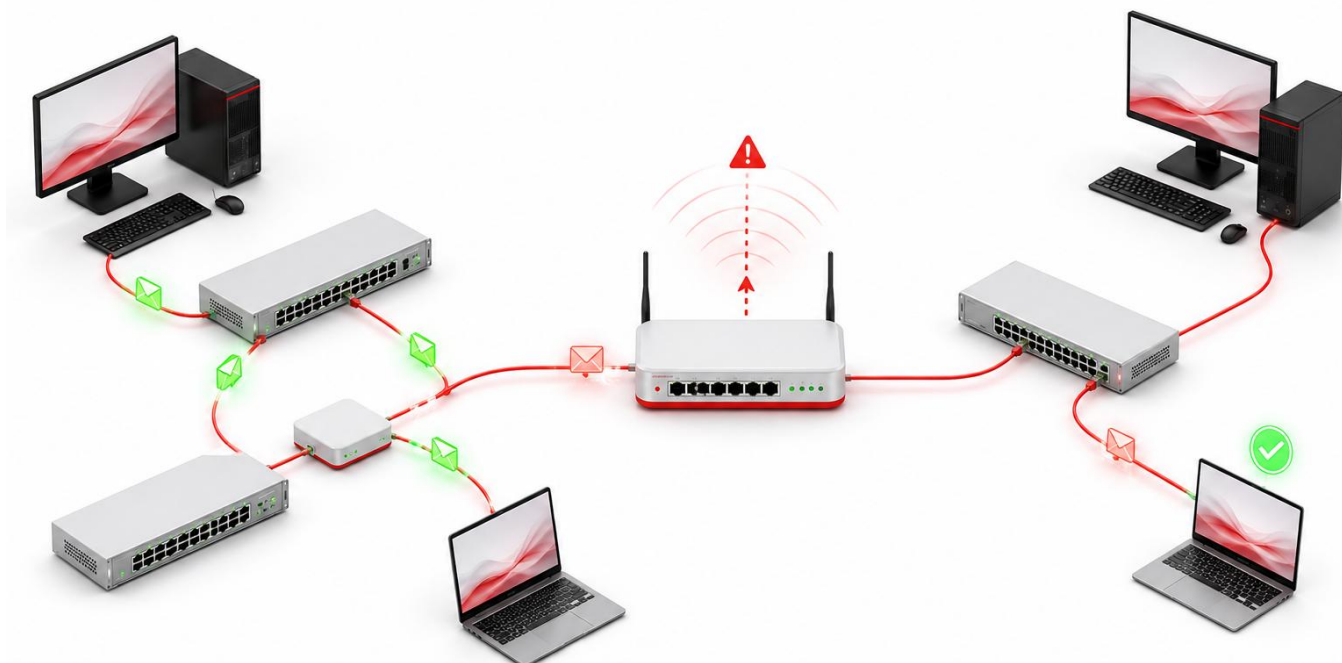
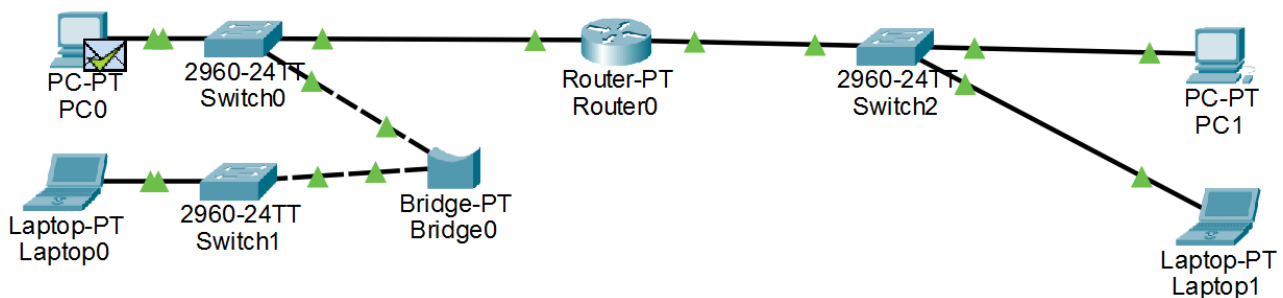
Grab the **closed yellow envelope** (Add Simple PDU). Click Town A's PC and send a message to Town A's Laptop.

It should say **Successful** immediately. The Bridge successfully passed the message between the two switches!

Test the Router

Grab a new envelope. Click Town A's Laptop and send a message to Town B's Laptop.

Remember: The first attempt across a router always says **Failed** while it draws a map. Double-click the red "Failed" text (or send a new envelope). The second try will say **Successful**!



Lab 9: Exploring Everyday Network Devices

In your previous labs, you used Switches, Bridges, and Routers to move data around. Now, we are going to look at **Network Devices** you use every single day in the real world: **WiFi Routers, Smartphones, and Servers**.

Cheat Sheet

Before we begin, get familiar with the three key devices you'll be working with in this lab.



The Server (The Digital Library)

A super-computer that holds websites, videos, or files, and "serves" them to anyone who asks.



The Home Router (The WiFi Box)

This is the box you have at home. It is actually **three devices in one!** It is a Switch (has ports on the back), a Router (connects to the internet), and an Access Point (gives out invisible WiFi signals).



Smart Device (Phones/Tablets)

Devices that have no cables and connect through the air.

Lab Outcomes

By the end of this lab, you will be able to:

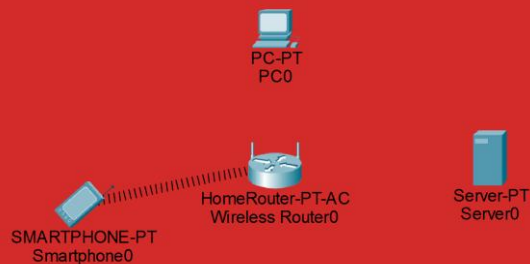
- Identify and place a Server, a Home WiFi Router, and a Smartphone in Packet Tracer.
- Understand the magic of **DHCP** (how devices get IP addresses automatically without you typing them).
- Connect a wireless device to a network without any cables.
- Use the graphical Web Browser on a virtual computer to view a website hosted on your Server.

Lab Work: Step-by-Step Guide

We are going to build a network that looks exactly like a normal house or a small coffee shop.

Step 1: Bring the New Devices onto the Screen

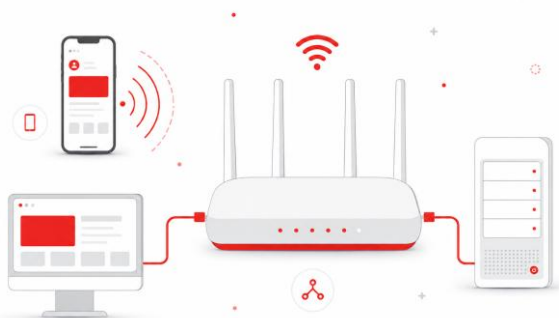
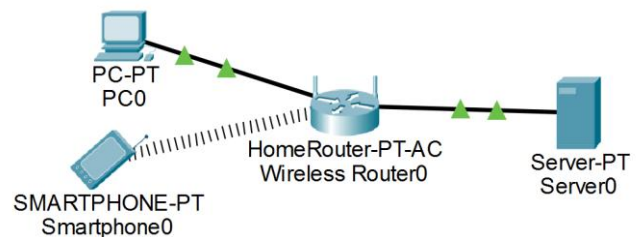
1. **The WiFi Box:** Go to **Network Devices** → **Wireless Devices**. Look for **Home Router** (or **WRT300N**) and place it in the centre of your screen.
2. **The Server:** Go to **End Devices**. Click the **Server** icon (tall computer tower) and place it on the left.
3. **The Wired PC:** Stay in **End Devices** and place a normal **PC** near the router.
4. **The Wireless Phone:** Place a **Smart Device** (Smartphone) on the right. *Look closely!* Within a few seconds, magical wavy lines should appear between the Smartphone and the Home Router – it connected to the WiFi all by itself!



Step 2: Plug in the Wired Devices

Let's plug our Server and our PC into the back of the Home Router.

1. Click the **Connections** icon (the orange lightning bolt).
2. Click the **Magic Auto-Connect Cable** (the yellow lightning bolt with sparkles).
3. Click the **Server** and then click the **Home Router**.
4. Click the **PC** and then click the **Home Router**. *Wait for the little orange dots on the cables to turn green!*



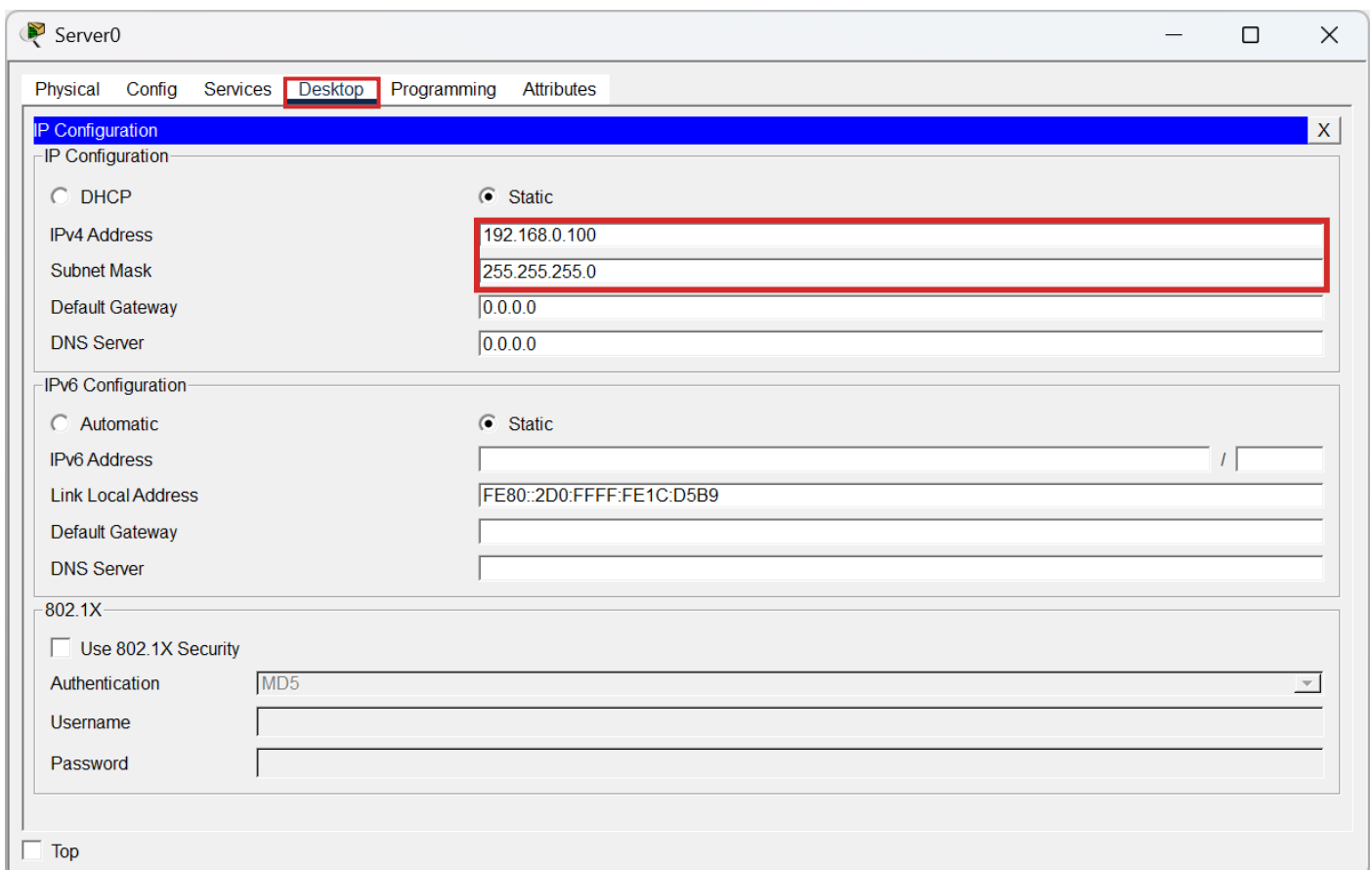
Configuration & Testing

1

Step 3: Setup the Server (The Website Host)

The server needs a permanent IP address so our devices know exactly where to find the website.

1. Click on the **Server**.
2. Go to **Desktop** tab → **IP Configuration**.
3. Type **192.168.0.100** in the **IPv4 Address** box.
4. Click the **Subnet Mask** box so it auto-fills (**255.255.255.0**).
5. Close the window.

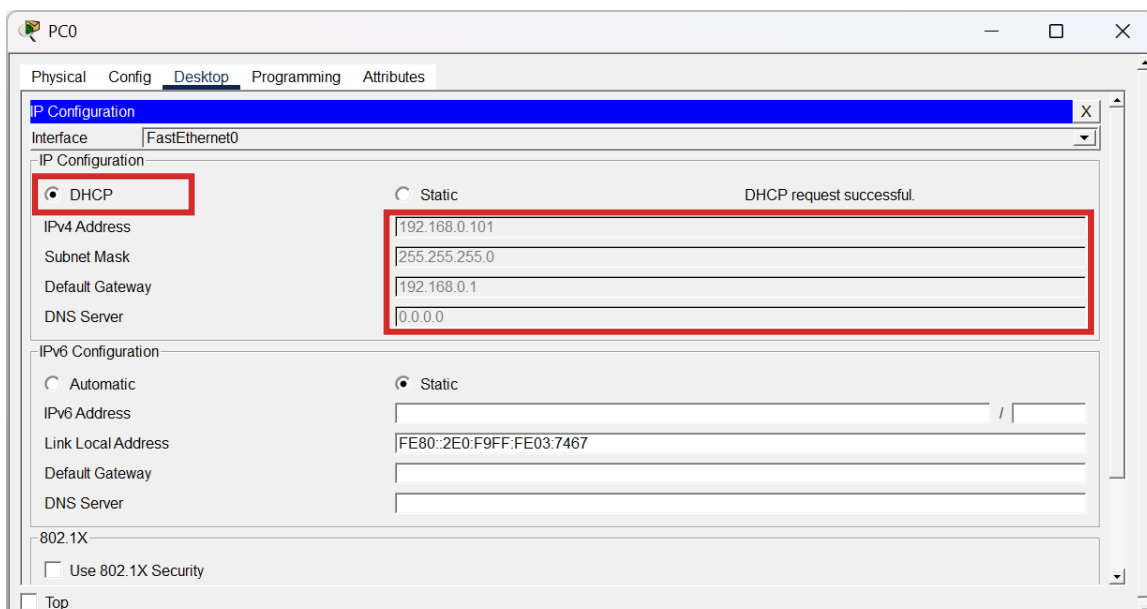
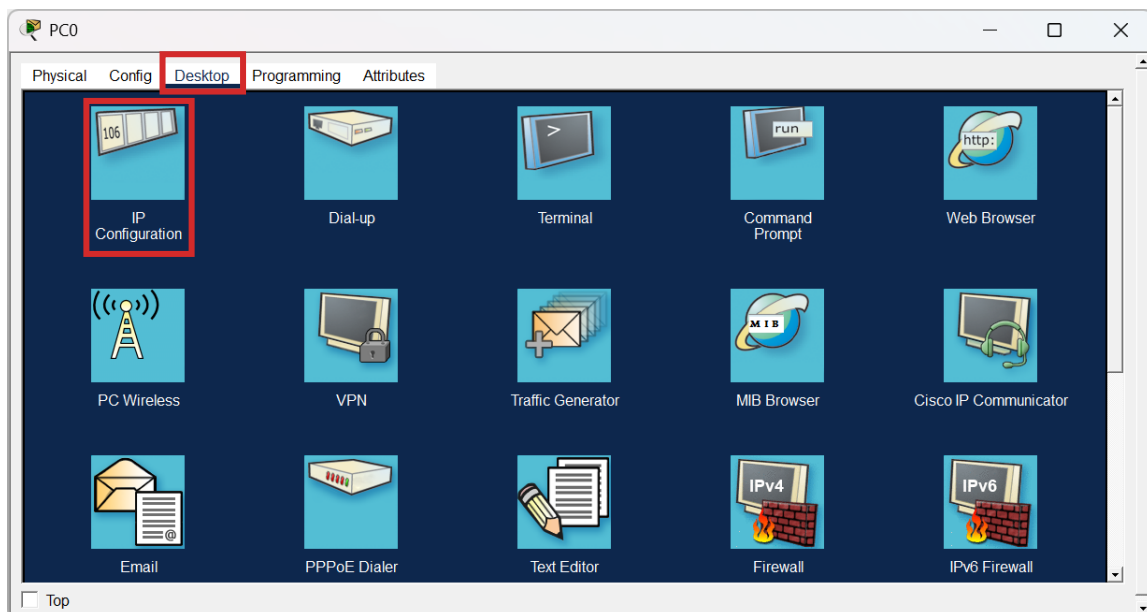


2

Step 4: The Magic of DHCP (Automatic IP Addresses)

In previous labs, you had to type the IP address for every single computer. In the real world, your Home Router hands them out automatically using a feature called **DHCP**! Let's turn it on.

1. Click on your **PC** → **Desktop** → **IP Configuration**.
2. Instead of typing numbers, click the little circle button next to **DHCP** (it defaults to Static).
3. Wait two seconds. You will see the boxes magically fill themselves with numbers (like **192.168.0.101**)! Next to it, it will say *"DHCP request successful."* Close the window.
4. Click on your **Smartphone** → **Desktop** → **IP Configuration**. Notice that DHCP is *already* selected, and it already gave itself an IP address (like **192.168.0.102**).

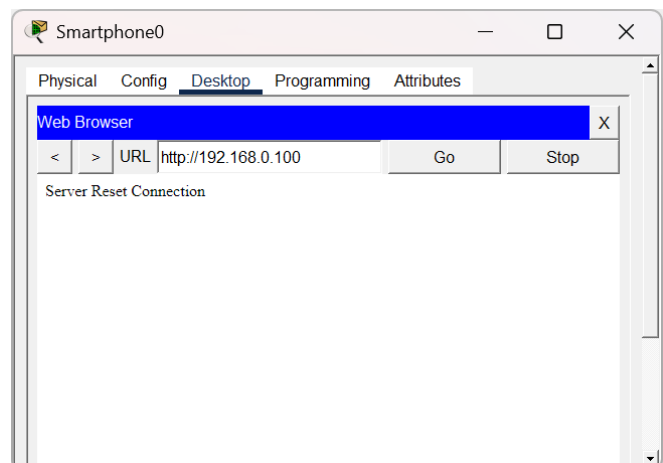
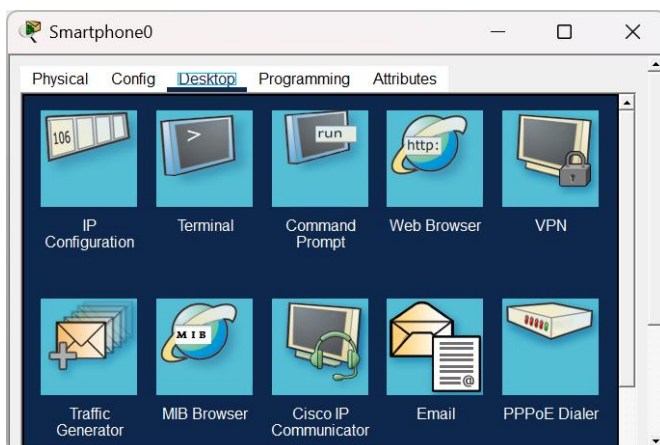
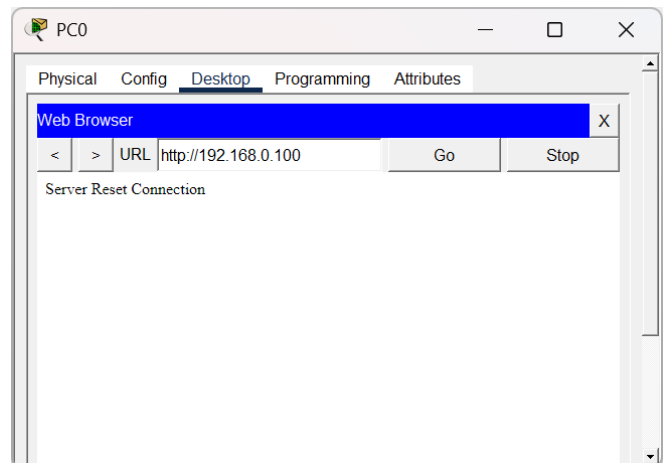
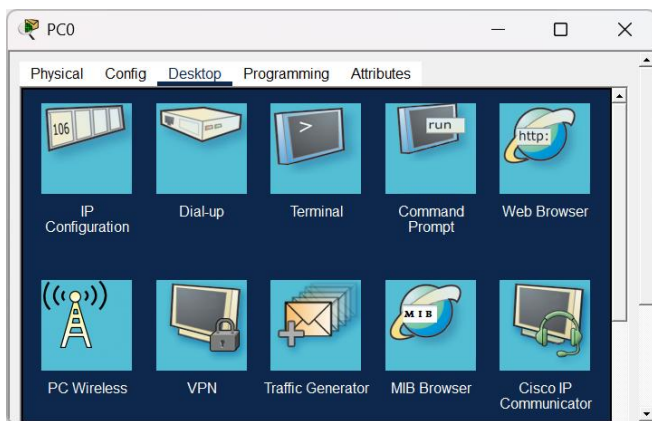


3

Step 5: Test the Network — Surf the Web!

Instead of sending a boring digital envelope, let's actually load a webpage from our Server!

1. Click on your **PC** → **Desktop** tab → click the **Web Browser** icon (little blue globe).
2. A blank web browser will open. In the URL bar at the top, type: **192.168.0.100**
3. Click **Go**. *Success!* You should see a blue screen that says "Cisco Packet Tracer" with welcome text. You just accessed a website!
4. **Bonus Test:** Close the PC window. Click on your **Smartphone**, open its **Web Browser**, and type **192.168.0.100**. Click **Go**. Your wireless phone can see the website too!



Congratulations! You have just set up a realistic, modern network featuring a Server, an automatic WiFi Router, and both wired and wireless devices.

Lab # 10: Static Routing

DRAWING THE MAP

In this lab, you will connect two routers together, understand why routers drop messages when they don't know the destination, use the GUI to configure a static route (manually teaching the router a new path), and successfully send a message across a multi-router network.

Connect Routers

Connect two different routers together

Understand Routing

Understand why routers drop messages when they don't know the destination

Configure Static Routes

Use the GUI to configure a "Static Route" – manually teaching the router a new path

Send Messages

Successfully send a message across a multi-router network

Lab Work: Building & Configuring the Network

Step 1: Build the World

Two towns, two post offices. Place **PC0**, a **2960 Switch**, and **Router0** on the left. Place **PC1**, a **2960 Switch**, and **Router1** on the right.

Step 2: Wire It Together

Use the **Auto-Connect Cable** (yellow lightning bolt):

1. PC0 → Left Switch
2. Left Switch → Router0
3. PC1 → Right Switch
4. Right Switch → Router1
5. Router0 → Router1 (*red cable is normal!*)

Step 3: Configure Router Ports

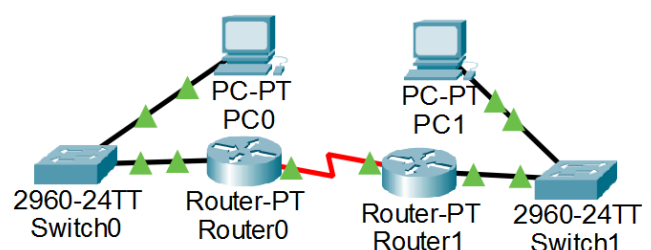
Router0 – FastEthernet0/0 (Town A): 192.168.1.1 / FastEthernet0/1 (to Router1): 10.0.0.1

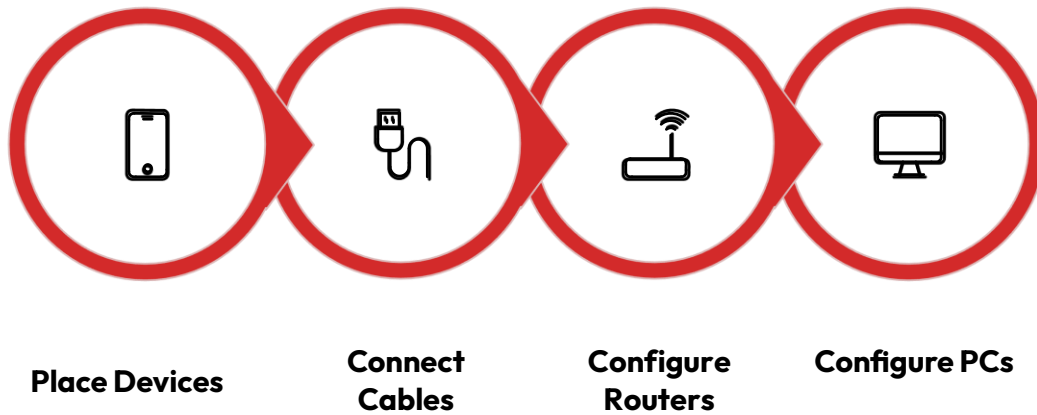
Router1 – FastEthernet0/0 (Town B): 192.168.2.1 / FastEthernet0/1 (to Router0): 10.0.0.2

Step 4: Configure the PCs

PC0: IPv4 192.168.1.10 · Gateway 192.168.1.1

PC1: IPv4 192.168.2.10 · Gateway 192.168.2.1





Once configured, the router-to-router cable should show **green dots** – your physical link is live!

Static Routing & The Final Test

Step 5: The Problem

Send a yellow envelope from PC0 → PC1. It says **Failed!** Router0 has no idea where 192.168.2.0 is – its map is blank. Let's draw the map.

Step 6: Give the Routers the Map

Router0 → Config → ROUTING → Static → Add:

- **Network:** 192.168.2.0 (Town B)
- **Mask:** 255.255.255.0
- **Next Hop:** 10.0.0.2 (Router1)

Router1 → Config → ROUTING → Static → Add:

- **Network:** 192.168.1.0 (Town A)
- **Mask:** 255.255.255.0
- **Next Hop:** 10.0.0.1 (Router0)

Step 7: The Final Test! 🎉

Grab a **brand new yellow envelope**. Click PC0, then PC1.

It might fail on the *very first try* while routers update their maps – send a second envelope and it will say **Successful!**

- ✅ Both routers now know exactly how to forward mail in both directions. The map is complete!

Lab 11: RIP Configuration

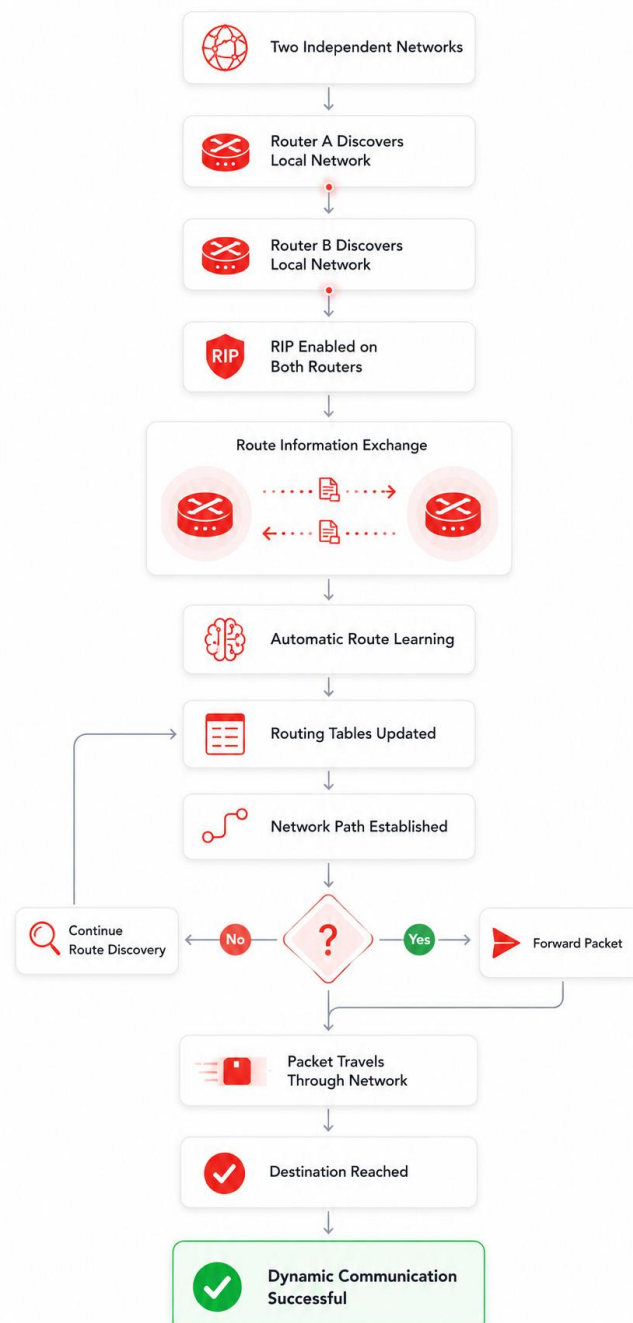
Routing Information Protocol – Learn how routers automatically share their maps and route messages across a network without manual configuration.

NETWORKING LAB

DYNAMIC ROUTING

By the End of This Lab

- Understand the difference between **Static** (manual) and **Dynamic** (automatic) routing
- Understand how RIP allows routers to share their maps automatically
- Use the GUI to enable RIP on a router
- Successfully send messages across a network without manually typing every destination path

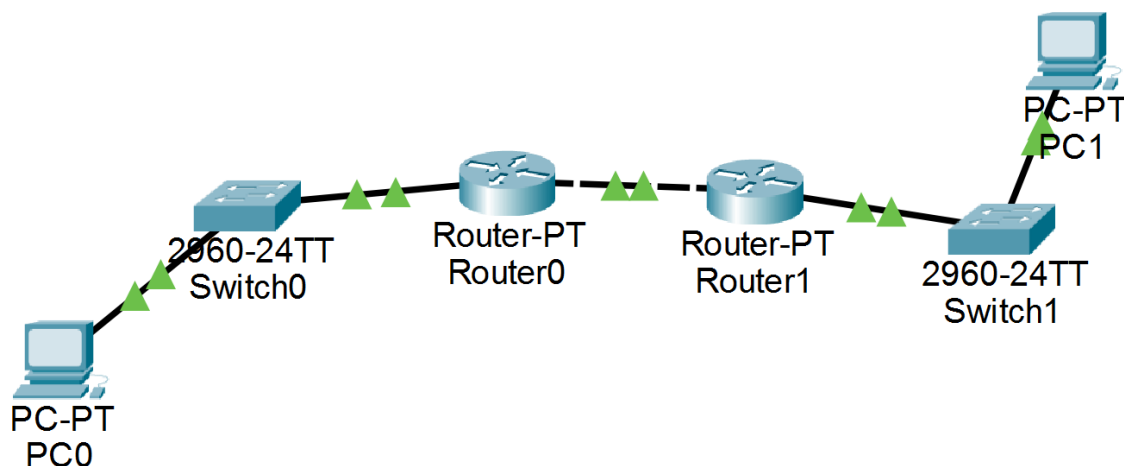


Building the Network

Step 1: Build the World

Place two towns and two routers to demonstrate how they share information.

- **Town A (Left):** PC0 → 2960 Switch → Router0
- **Town B (Right):** PC1 → 2960 Switch → Router1
- Use the **Magic Auto-Connect Cable** to link Router0 ↔ Router1 (the highway)



Step 2: Configure IP Addresses

Give every device its address. You're a pro at this by now!

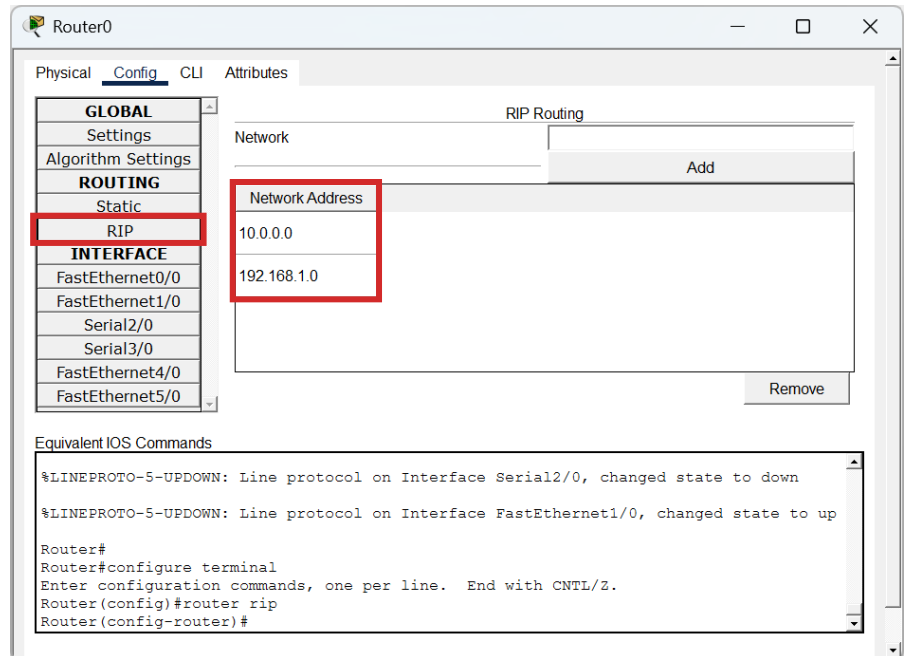
Device	Interface	IP Address	Gateway
Router0 (Left)	FastEthernet0/0	192.168.1.1	—
Router0 (Left)	FastEthernet0/1	10.0.0.1	—
Router1 (Right)	FastEthernet0/0	192.168.2.1	—
Router1 (Right)	FastEthernet0/1	10.0.0.2	—
PC0	Desktop → IP Config	192.168.1.10	192.168.1.1
PC1	Desktop → IP Config	192.168.2.10	192.168.2.1

Step 3: The Magic of RIP

- Golden Rule of RIP:** You only tell the router about networks **directly plugged into it**. The router figures out the rest!

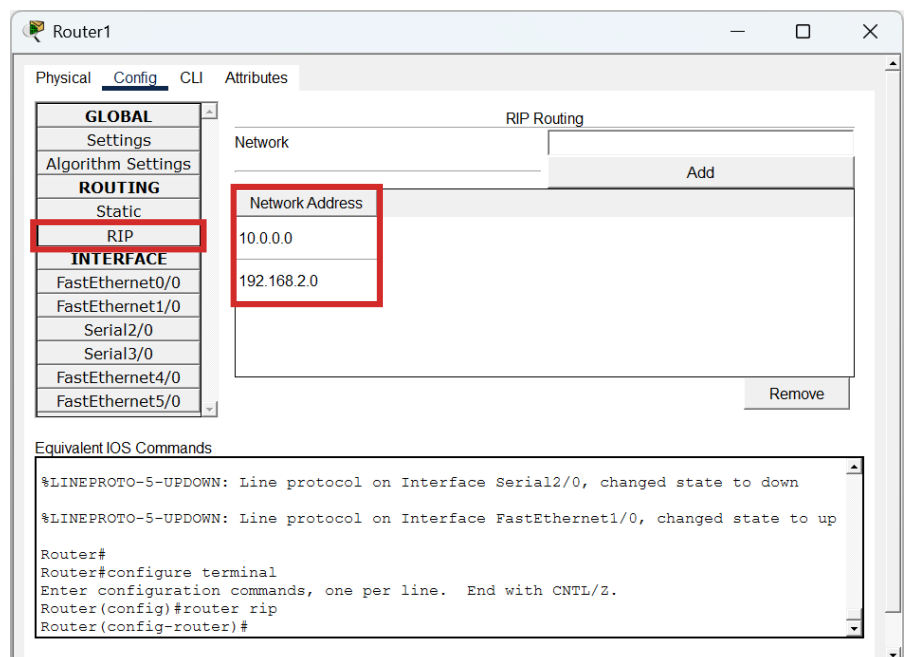
Configure RIP on Router0

1. Click **Router0** → **Config** tab
2. Under **ROUTING**, click **RIP**
3. Type **192.168.1.0** → click **Add**
4. Type **10.0.0.0** → click **Add**
5. Close the window



Configure RIP on Router1

1. Click **Router1** → **Config** tab
2. Under **ROUTING**, click **RIP**
3. Type **192.168.2.0** → click **Add**
4. Type **10.0.0.0** → click **Add**
5. Close the window



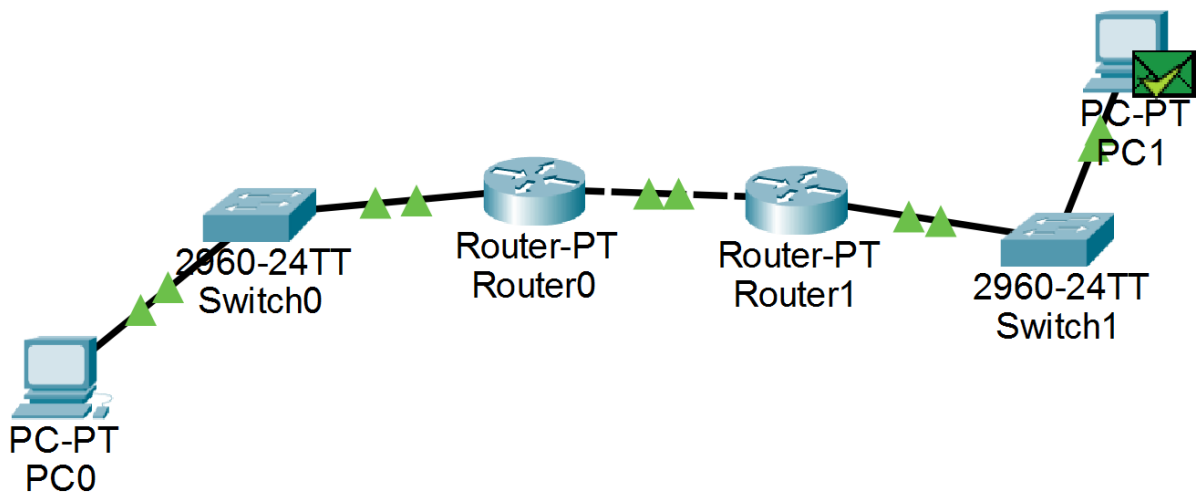
Step 4: Wait for the Gossip

In the real world, RIP sends out its maps every **30 seconds**. In Packet Tracer it usually happens instantly – but give the routers **3–5 seconds** to "talk" behind the scenes.

Router 0 is telling Router 1: *"Hey! I know how to get to 192.168.1.0!"* – and Router 1 saves that to its map.

Step 5: Test the Automatic Map

1. Grab the **closed green envelope** (Add Simple PDU)
2. Click **PC0** (Sender in Town A)
3. Click **PC1** (Receiver in Town B)
4. First attempt may say **Failed**, routers are preparing the path
5. Double-click the red "Failed" text or send a new envelope
6. It will say **Successful!** ✓



Congratulations! You didn't have to tell Router 0 how to find Town B. It learned the map all by itself just by listening to Router 1. You have successfully configured **dynamic routing**!

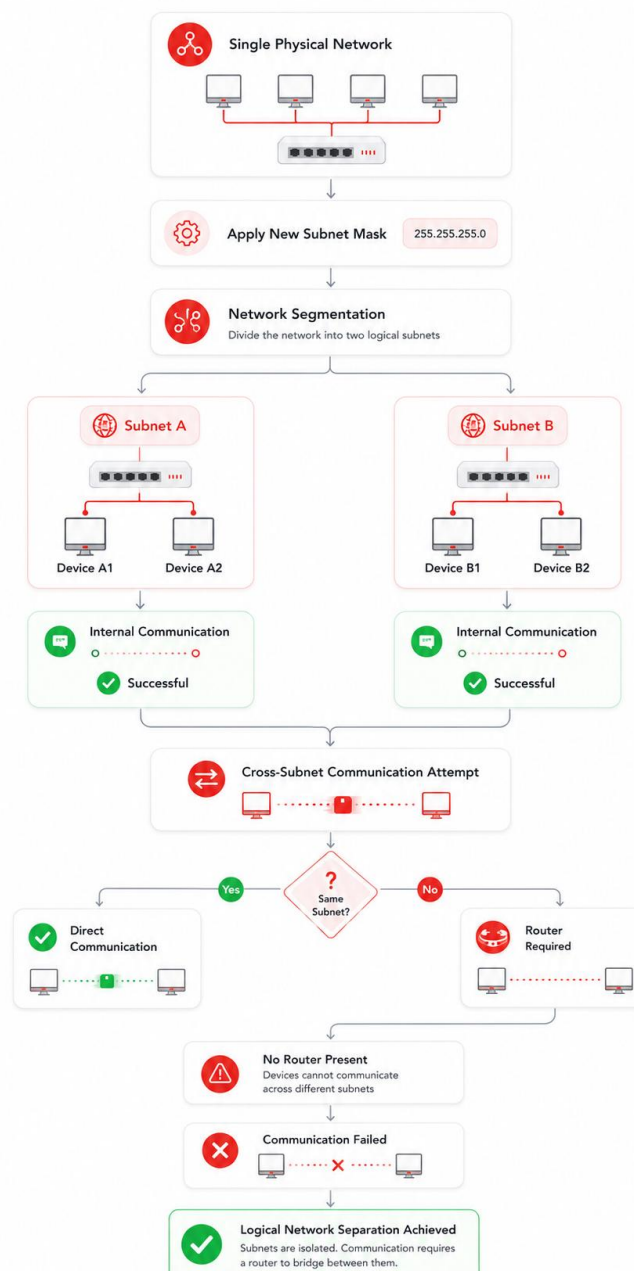
Lab 12: Introduction to Subnetting

✂ SLICING THE NETWORK

By the end of this lab, you will understand how a Subnet Mask works, change it in a graphical interface, logically divide a single physical network into two separate subnets, and prove that computers on different subnets cannot communicate without a router – even when plugged into the same switch.

By the end of this lab, you will be able to:

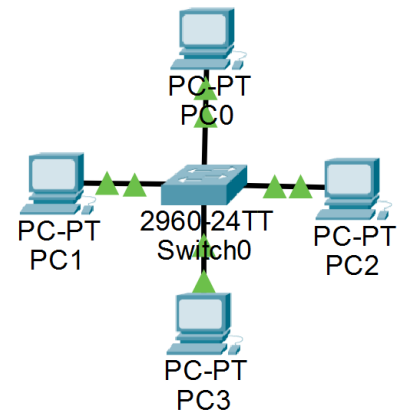
- Understand what a **Subnet Mask** does
- Change the default Subnet Mask in the graphical interface
- Logically divide a single physical network into two separate **Subnets**
- Prove that computers on different subnets **cannot talk to each other** without a router, even if they are plugged into the same switch



Step 1: Build the Physical Network

Let's build a very simple, single network.

1. Go to **End Devices** and place **four PCs** on your screen (PC0, PC1, PC2, PC3)
2. Go to **Network Devices** → **Switches** and place one **2960 Switch** in the middle
3. Grab the **Magic Auto-Connect Cable** (yellow lightning bolt with sparkles)
4. Connect all four PCs to the switch – *wait for all the little orange dots on the cables to turn green!*



Steps 2 & 3: Configure Subnet A and Subnet B

We are going to change the Subnet Mask from 255.255.255.0 to 255.255.255.128. By putting a **128** at the end, we slice the network exactly in half.

Subnet A — The Left Side

Gets addresses from 1 to 126

PC0 and PC1 go here

Subnet B — The Right Side

Gets addresses from 129 to 254

PC2 and PC3 go here

Configure Subnet A (PC0 & PC1)

1. Click PC0 → **Desktop** tab → **IP Configuration**
2. **IPv4 Address:** Type 192.168.1.10 (*10 is smaller than 126 → Subnet A*)
3. **Subnet Mask:** Click the box, delete the 0 and change it to 128 → 255.255.255.128. Close the window.
4. Click PC1 → **Desktop** → **IP Configuration**
5. **IPv4 Address:** Type 192.168.1.20
6. **Subnet Mask:** Delete the 0 and change it to 128. Close the window.

Configure Subnet B (PC2 & PC3)

1. Click PC2 → **Desktop** → **IP Configuration**
2. **IPv4 Address:** Type 192.168.1.150 (*150 is larger than 128 → Subnet B*)
3. **Subnet Mask:** Click the box, delete the 0, and change it to 128. Close the window.
4. Click PC3 → **Desktop** → **IP Configuration**
5. **IPv4 Address:** Type 192.168.1.160
6. **Subnet Mask:** Delete the 0 and change it to 128. Close the window.

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.10

Subnet Mask 255.255.255.128

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::260:5CFF:FE52:D947

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC1

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.20

Subnet Mask 255.255.255.128

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::20C:CFFF:FEA1:B429

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC2

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.150

Subnet Mask 255.255.255.128

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::230:F2FF:FE1B:B867

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

PC3

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.160

Subnet Mask 255.255.255.128

Default Gateway 0.0.0.0

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address

Link Local Address FE80::207:ECFF:FEED:6839

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

☐ Top

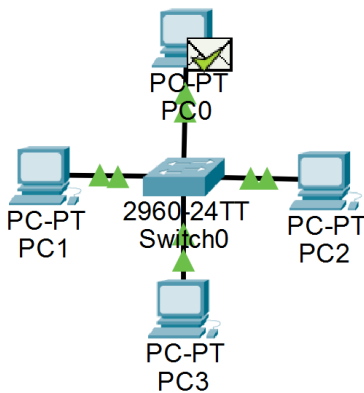
Step 4: Testing the Subnets

Even though all four computers are plugged into the **exact same physical switch**, their digital addresses have separated them into two private clubs. Let's prove it!

1

Test Subnet A

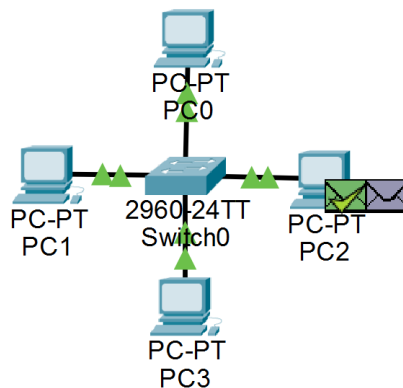
Grab the **closed yellow envelope** (Add Simple PDU) from the top right. Click **PC0** and send it to **PC1**. Look at the bottom right – it says **Successful!** They are in the same private club.



2

Test Subnet B

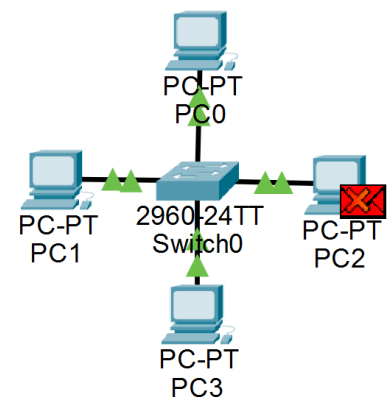
Grab a new envelope. Click **PC2** and send it to **PC3**. It says **Successful!** They are in their own private club.



3

The Ultimate Test

Grab a new envelope. Click **PC0** (Subnet A) and try to send it to **PC2** (Subnet B). It says **Failed!** Try it as many times as you want – it will *always* fail.



Why did it fail? Because PC0 looked at its Subnet Mask (**128**) and realised that PC2 (**150**) lives in a completely different neighbourhood. To talk to a different neighbourhood, you would need a **Router** (The Post Office), just like you learned in Lab 6!



Congratulations! You just performed your first Subnetting task, digitally slicing a network into smaller, secure pieces *without needing extra physical equipment*.

Lab 13: IoT Simulation

BUILDING A SMART HOME

Up until now, you have been connecting computers and routers. But today, refrigerators, door locks, and even lightbulbs connect to the internet. This is called the **Internet of Things (IoT)**.

In this lab, we are going to build a tiny smart home and control physical objects using a digital smartphone app!

Key Concepts

IoT (Internet of Things): Taking a normal, "dumb" object (like a ceiling fan) and putting a tiny WiFi computer inside it so you can control it from your phone.

The Home Gateway: Think of this as the "brain" of your smart home. It acts as the central hub that all your smart devices talk to.

Lab Outcomes

By the end of this lab, you will be able to:

- Understand the concept of the Internet of Things (IoT)
- Locate and place Smart Home devices into the Packet Tracer workspace
- Connect IoT devices to a central Home Gateway hub
- Use a Smartphone's "IoT Monitor" app to remotely control physical objects (like turning on a light or a fan)

Lab Work: Steps 1–3

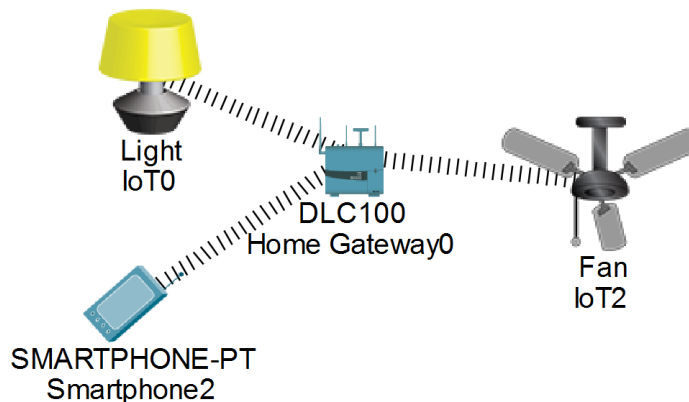
Buy the Smart Devices

Plug in the Smart Objects

Register the Devices to the Brain

Let's set up a room with a smart fan and a smart lightbulb, plus a central hub to control them.

1. **The Brain:** Go to **Network Devices** → **Wireless Devices**. Look for the **Home Gateway** and place it in the center of your screen.
2. **The Smart Objects:** Go to **End Devices**. Look at the sub-menu below it and click on the **Home** icon (it looks like a little house).
3. Find the **Smart Light** and place it on your screen.
4. Find the **Smart Fan** (or Smart Door) and place it on your screen.
5. **The Remote Control:** Go back to the main **End Devices** list (the computer icon) and place a **Smartphone** on the screen. *(It will magically connect to the Home Gateway's WiFi right away with wavy lines!)*



1. Click on the **Smart Light**.
2. Go to the **Config** tab.
3. Look at the very bottom of the left-hand menu for a section called **IoT Server**.
4. Click the little circle next to **Home Gateway**. (This tells the light, "Listen to the Home Gateway!"). Close the window.
5. Click on the **Smart Fan**.
6. Go to the **Config** tab.
7. Under **IoT Server**, click the circle next to **Home Gateway**. Close the window.

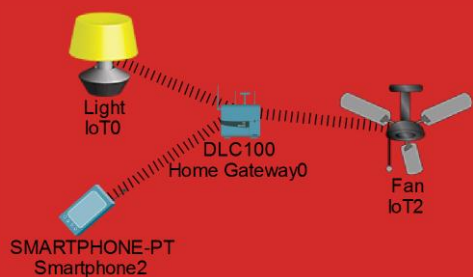
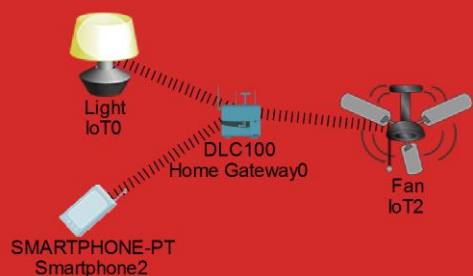
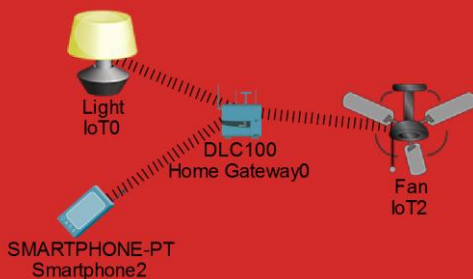
Lab Work: Steps 4–5 & Pro Tips

Bonus Pro-Tip: Manual Override

If you want to walk up to the smart device and physically turn it on without your phone, hold down the **Alt key** on your keyboard (or **Option key** on a Mac) and click directly on the fan or lightbulb on your screen. This simulates manually flipping the switch!

Congratulations!

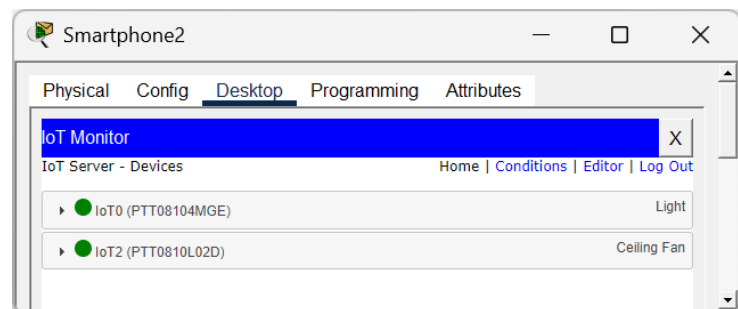
You have successfully built and configured your very first **Internet of Things network**.



1

Open the Smartphone App

1. Click on your **Smartphone**.
2. Go to the **Desktop** tab.
3. Look for an icon called **IoT Monitor** (it usually has a little house or gear on it) and click it.
4. A login screen will pop up. Because we are using the simple Home Gateway, you don't even need a password. Just click the **Login** button!



2

Control Your Smart Home!

If you did everything right, your IoT Monitor app will now show a list with two items on it (usually named IoT Device 0 and IoT Device 1).

1. Click on the first device in the list (The Smart Light).
2. You will see buttons that say **Dim**, **On**, and **Off**.
3. Click **On**. Now, move the smartphone window over so you can see your main workspace. The lightbulb on your screen has actually turned yellow and is shining!
4. Click on the second device in the list (The Smart Fan).
5. You will see buttons for **Low**, **High**, and **Off**.
6. Click **High**. Look at your workspace—the fan blades are now spinning with action lines!

Lab 14: Configuring VoIP Phones

 IP TELEPHONY

PACKET TRACER

Up until now, we have been sending digital "envelopes" between computers. But what if we want to send **human voices**?

For this lab:



VoIP (Voice over IP)

This stands for making phone calls over the internet (or a computer network) instead of using old-school telephone wires.



IP Phones

These look like normal office phones, but they plug into a computer Switch instead of a phone jack!



The Secret Code

Packet Tracer does not have a graphical menu for telephones. So, for the very first time, we are going to use the **Command Line Interface (CLI)** to type in some "cheat codes" to make the router act like a telephone company!

Lab Outcomes

By the end of this lab, you will be able to:

1

Understand VoIP

Understand what VoIP means.

2

Power the Phones

Physically plug in the power adapters for IP Phones inside Packet Tracer.

3

Use the CLI

Use basic command-line codes to turn a Router into a telephone switchboard.

4

Assign Extensions

Assign phone numbers (extensions) to IP phones.

5

Make a Call

Successfully dial one IP phone from another and hear it ring!

Step 1: The Hardware Setup

We must use a specific router for this because only certain routers are smart enough to handle phone calls.

01

Add the Router

Go to **Network Devices** → **Routers**. Look for the **2811 Router** and place it at the top of your screen.

02

Add the Switch

Go to **Network Devices** → **Switches**. Place a **2960 Switch** in the middle.

03

Add the IP Phones

Go to **End Devices**. Click on the **IP Phone** icon (it looks like an office telephone) and place **two** of them at the bottom (IP Phone 0 and IP Phone 1).



Step 2: Plug in the Phones!



Warning: IP Phones in Packet Tracer are turned OFF by default. You have to physically plug them into the wall!

1 Open IP Phone 0

Click on IP Phone 0.

2 Stay on the Physical Tab

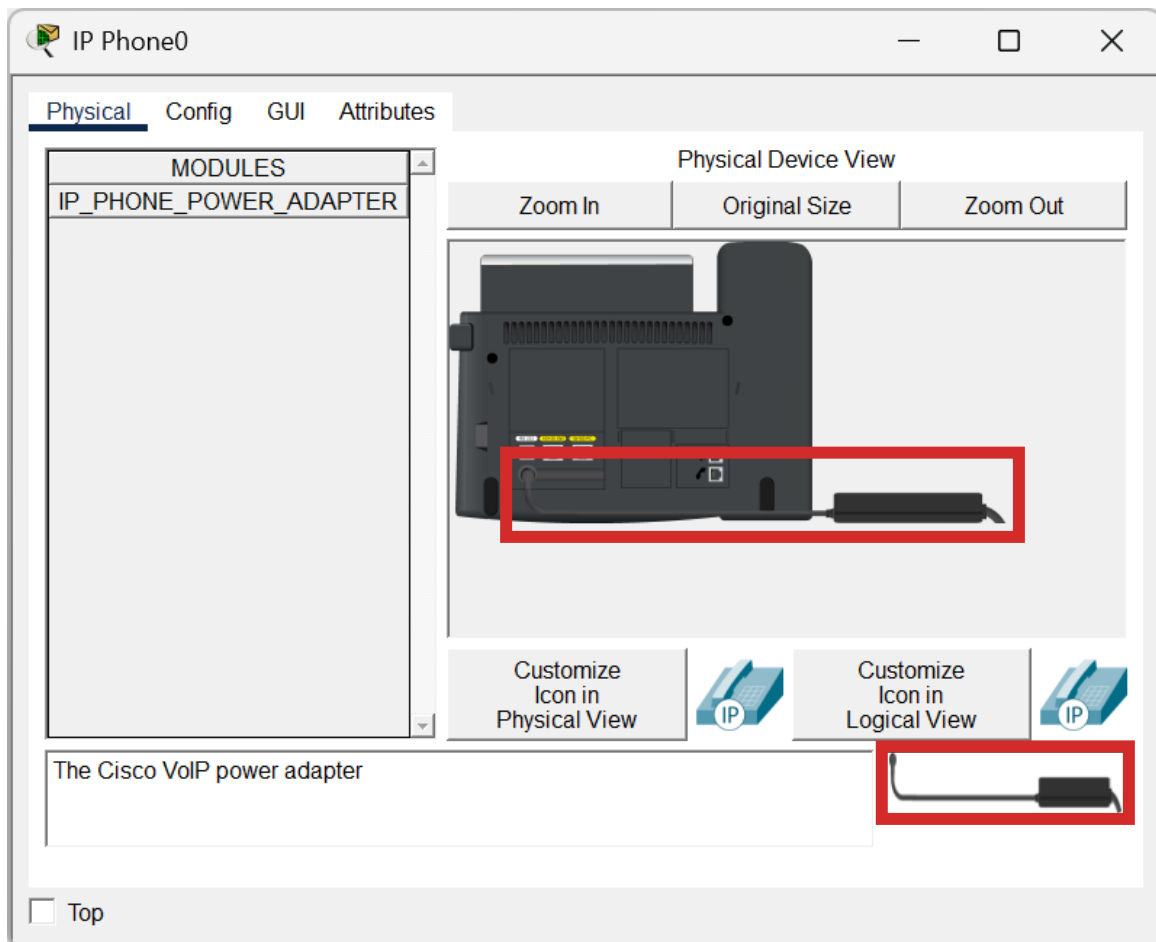
Look closely at the picture of the phone; there is a black power adapter sitting in the bottom right corner.

3 Drag the Power Adapter

Click and drag that black power adapter and drop it onto the little circular power plug hole on the phone. (You will know it worked if you see the cord plugged in).

4 Repeat for IP Phone 1

Close the window and **do the exact same thing** for IP Phone 1.



Step 3: Wire It Up

→ **Grab the Magic Auto-Connect Cable**

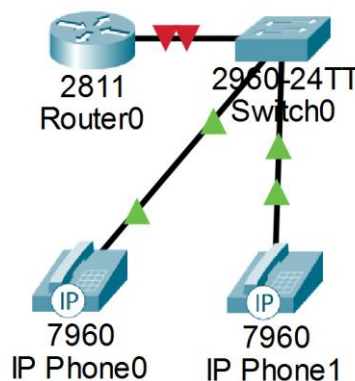
Use the **yellow lightning bolt** cable.

→ **Connect IP Phone 0 → Switch**

→ **Connect IP Phone 1 → Switch**

→ **Connect Switch → 2811 Router**

Wait for the dots on the phones to turn green!



Step 4: Program the Switch (The Command Line)

We have to tell the switch that it is allowed to carry voice traffic.

01

Open the Switch CLI

Click on the **Switch**. At the top, click on the **CLI** tab (Command Line Interface). Press **Enter** on your keyboard to get started. You will see **Switch>**.

02

Type the Commands

Type these codes exactly as written, pressing **Enter** after every single line:

```
enable
configure terminal
interface range fa0/1 - 5
switchport voice vlan 1
```

Close the Switch window.

Step 5: Program the Router (The Telephone Company)

Now we must give the router its IP address, set up DHCP (so the phones get IP addresses automatically), and tell it to handle phone calls.

Turn on the Router's Gate

Set up DHCP so phones get an IP address automatically

Turn on Telephone Services and assign phone numbers

Give Phone 1 a phone number

Give Phone 2 a phone number

Click on the **2811 Router**. Go to the **CLI** tab. If it asks *"Continue with configuration dialog? [yes/no]"*, type **no** and press Enter. Press **Enter** again to get started. Type these codes exactly, pressing **Enter** after every line:

```
enable
configure terminal
interface fa0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit
ip dhcp pool VOICE
network 192.168.1.0 255.255.255.0
default-router 192.168.1.1
option 150 ip 192.168.1.1
exit
telephony-service
max-ephones 2
max-dn 2
ip source-address 192.168.1.1 port 2000
auto assign 1 to 2
exit
ephone-dn 1
number 1111
ephone-dn 2
number 2222
```

Close the Router window.

Step 6: Make a Phone Call!

Let's see if your network can make a real phone call.

1 Check IP Phone 0's Number

Hover your mouse over **IP Phone 0**. You should see a little popup box that says its line number is **1111**.

2 Check IP Phone 1's Number

Hover over **IP Phone 1**. It should say its line number is **2222**.

3 Open Both GUI Tabs

Click on **IP Phone 0**. Go to the **GUI** tab. You will see a picture of the actual phone screen and buttons! Click on **IP Phone 1** and open its **GUI** tab too. Arrange the two windows side-by-side so you can see both phones.

4 Dial 2222

On **IP Phone 0**'s screen, use your mouse to click the keypad buttons and dial **2222**.

5 Watch It Ring!

Look at **IP Phone 1**. The red light will be flashing, and the screen will say **"Ring Out"** or **"From 1111"**!

6 Answer the Call

On **IP Phone 1**, click the physical phone receiver (the part you hold to your ear) to answer the call. The screens will now say **Connected!**





Lab 15: ACL Configuration

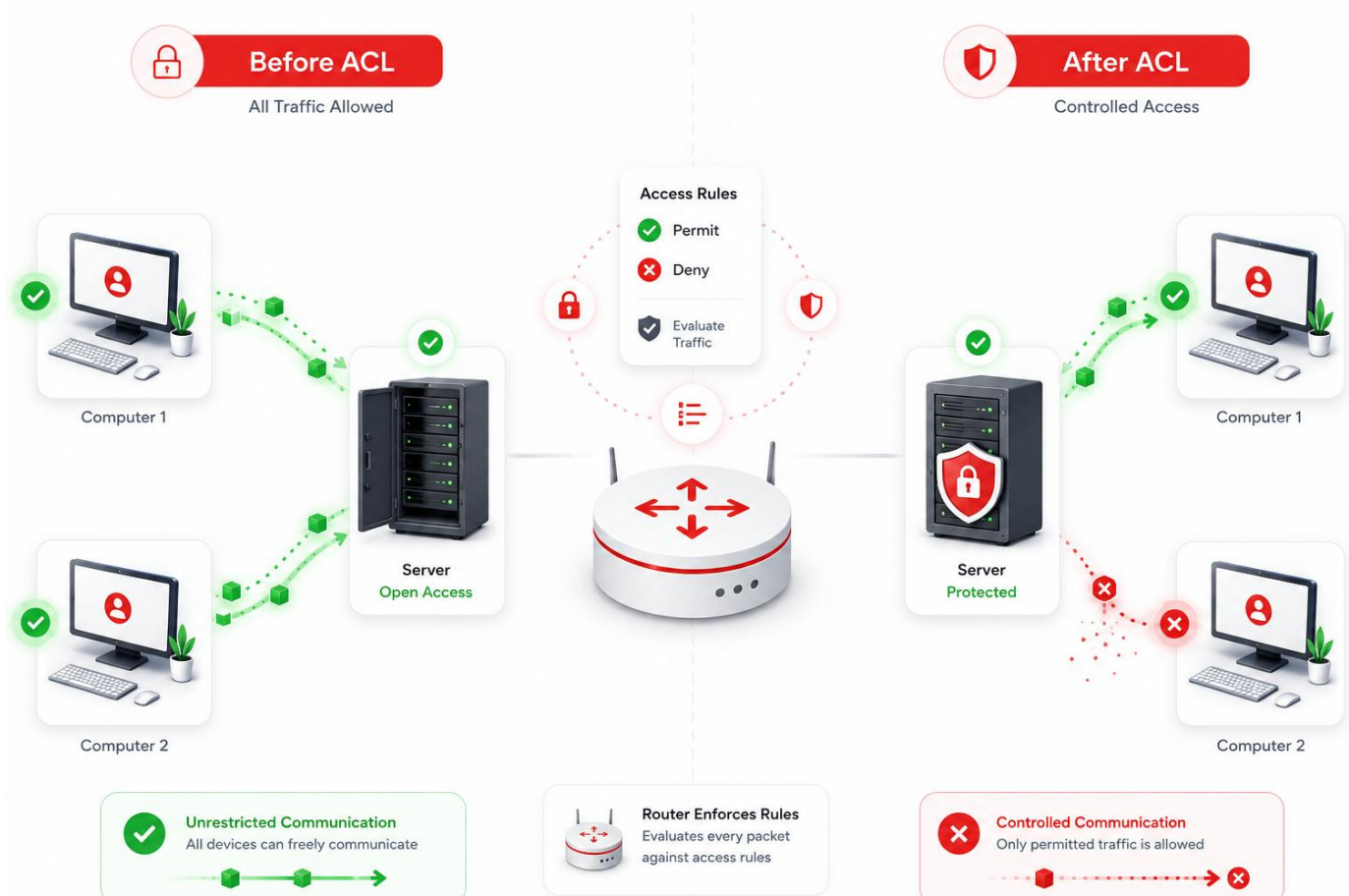
HIRING A SECURITY GUARD

Access Control Lists (ACLs) are the security guards of networking – they decide who gets in and who gets turned away. In this lab, you will build a network with a "Public" side and a "VIP" side, write security rules using the command line, and prove your ACL works by watching one computer succeed while another gets blocked.

Lab Outcomes & Network Setup

By the end of this lab, you will be able to:

- Understand what an Access Control List (ACL) does
- Build a network with a "Public" side and a "VIP" side
- Use the command line to create a security rule that bans a specific computer
- Apply the security guard to a specific router gate
- Prove the ACL works by watching one computer succeed and another get blocked



Step 1: Build the Network — The Club vs. The VIP Room

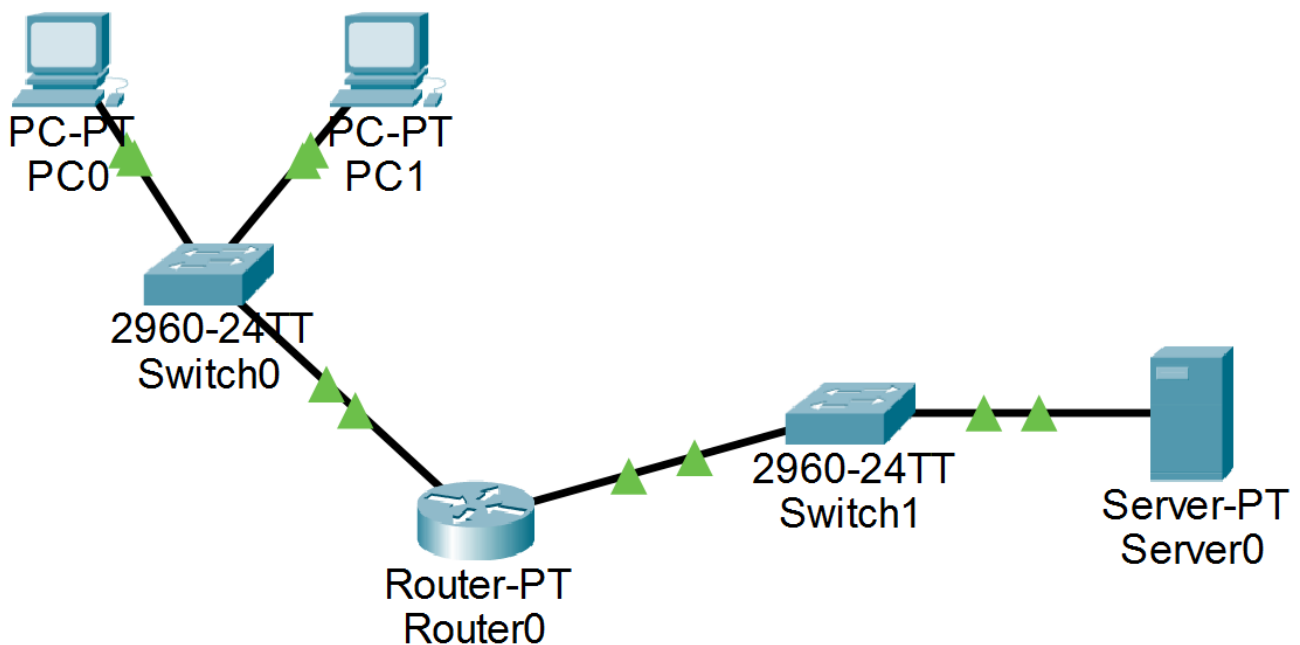
We are going to build a network where **PC0** is a good guy, **PC1** is a bad guy, and the **Server** is a highly secure VIP room.

The Public Side (Left): Place **two PCs** (**PC0** and **PC1**) and one **2960 Switch** on the left side of your screen.

The VIP Side (Right): Place one **Server** and one **2960 Switch** on the right side of your screen.

The Doors (Middle): Place a **PT-Router** in the center.

Step 2: Wire it Up – Grab the **Magic Auto-Connect Cable** (yellow lightning bolt) and connect: **PC0** → Left Switch, **PC1** → Left Switch, Left Switch → Router, **Server** → Right Switch, Right Switch → Router.



Configure the Router & Devices

01

Step 3: Open the Router Gates (Using the GUI)

Click the **Router** → **Config** tab.

02

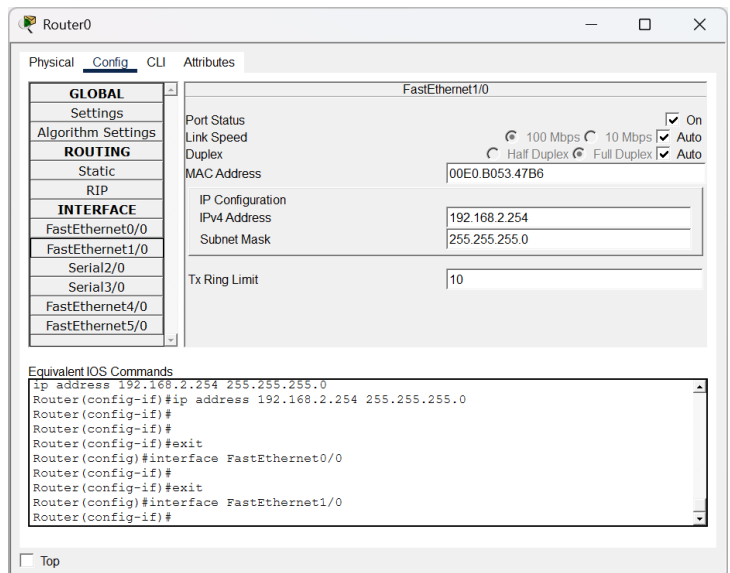
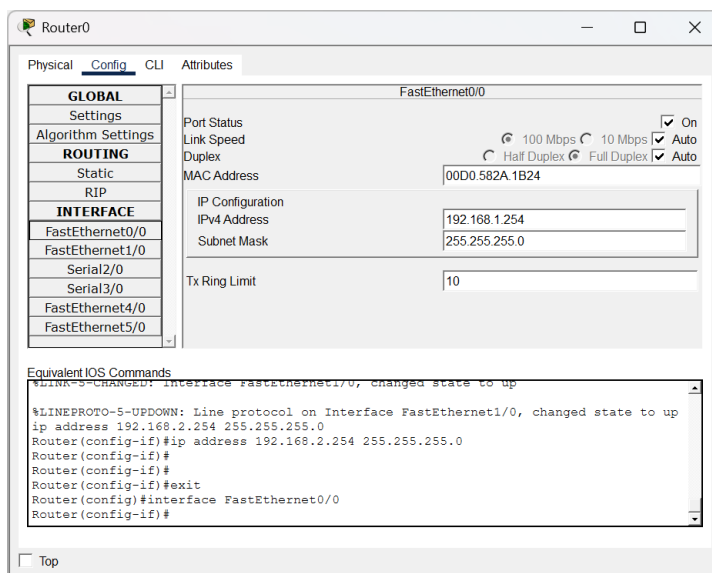
FastEthernet0/0 — Left Gate

Turn **Port Status: On**. IPv4 Address: **192.168.1.254** (Click Subnet to auto-fill).

03

FastEthernet0/1 — Right Gate

Turn **Port Status: On**. IPv4 Address: **192.168.2.254** (Click Subnet to auto-fill). Close the window.



PC0

Desktop → IP Config

IPv4: 192.168.1.1

Default Gateway:

192.168.1.254

PC1

Desktop → IP Config

IPv4: 192.168.1.2

Default Gateway:

192.168.1.254

Server

Desktop → IP Config

IPv4: 192.168.2.1

Default Gateway:

192.168.2.254

